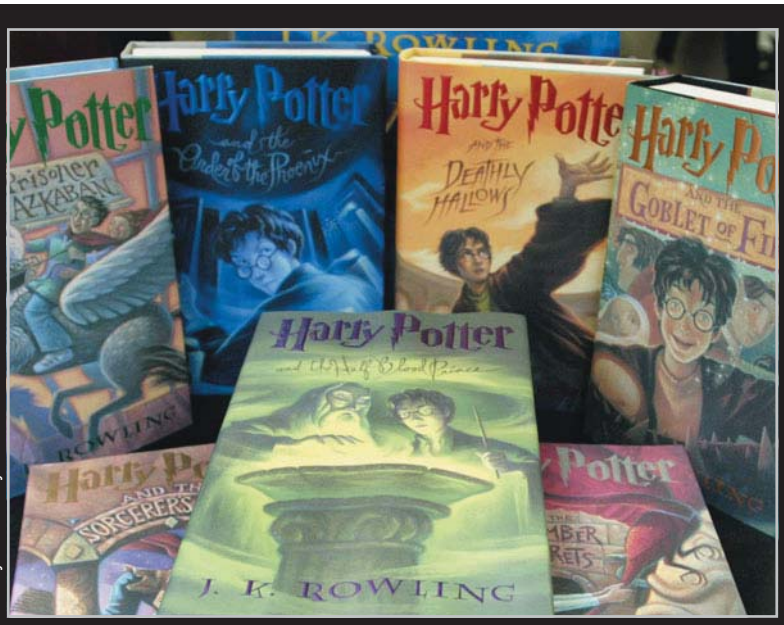


Bizuayehu Tesfaye/AP Photos

1

USING OPERATIONS TO COMPETE

The seventh novel in the Harry Potter series was released on July 21, 2007 and became an instant best seller around the globe. Because the book had to be delivered in a tight time window to the customers, Scholastic coordinated its publishing and distribution processes in USA months in advance of the release date.

Scholastic and Harry Potter

Scholastic is the world's largest publisher and distributor of children's books and educational materials. Founded in 1920, it had $1.9 billion in revenues in fiscal 2011 with offices in 16 countries including North America, Europe, Southeast Asia, Latin America, the Middle East, Australia, New Zealand, and Africa. Scholastic started planning in early 2007 for the worldwide release of the eagerly awaited seventh book *Harry Potter and the Deathly Hallows* in the acclaimed series by J.K. Rowling on the boy wizard. When the author finished the book in spring 2007, Scholastic's printers R.R. Donnelly & Sons and Quebecor World worked around the clock to make sure that the book would be ready by the release date. To save time in loading and unloading, Scholastic bypassed its own warehouses and required its truckers, Yellow Transportation and JB Hunt Transport Services, to use the same size trailers and pallets to ship books directly from six printing sites to big retailers like Barnes & Noble and Amazon.com. This fleet of trucks, if lined up bumper-to-bumper, would stretch for 15 miles. GPS transponders were used to alert Scholastic by e-mail if the driver or the trailer veered off the designated routes. The timing was particularly tricky for e-tailers, who had to directly ship books in advance for individual orders to arrive simultaneously around the country in order to minimize the risk of someone leaking the book's ending.

Since close to 90 percent of sales of such special books occur in the first week, they get special treatment to save time, money, space, and work. Scholastic had to customize, coordinate, and synchronize its operations and supply chain processes across multiple partners at the printing, warehousing, distribution, and retailing locations to ensure that the last book in the Harry Potter series reached the final customers no more than a few hours before the scheduled July 21, 12:01 A.M. release deadline. Not bad for a bunch of Muggles who transported 12 million copies in a short time window without the magical floo powder, portkeys, and broomsticks!

Source: Dean Foust, "Harry Potter and the Logistical Nightmare," *Business Week* (August 6, 2007), p. 9; Michelle Regenold, "Shipping Harry Potter: How Do They Do That?" **www.go-explore-trans.org/2007/mar-apr/shipping_HP.cfm; www.scholastic.com,** 2011.

LEARNING GOALS *After reading this chapter, you should be able to:*

1. Describe operations and supply chains in terms of inputs, processes, outputs, information flows, suppliers, and customers.
2. Define an operations strategy and its linkage to corporate strategy, as well as the role it plays as a source of competitive advantage in a global marketplace.
3. Identify nine competitive priorities used in operations strategy, and their linkage to marketing strategy.
4. Explain how operations can be used as a competitive weapon.
5. Identify the global trends and challenges facing operations management.

operations management
The systematic design, direction, and control of processes that transform inputs into services and products for internal, as well as external, customers.

process
Any activity or group of activities that takes one or more inputs, transforms them, and provides one or more outputs for its customers.

operation
A group of resources performing all or part of one or more processes.

supply chain
An interrelated series of processes within and across firms that produces a service or product to the satisfaction of customers.

supply chain management
The synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services, and information with customer demand.

Operations management refers to the systematic design, direction, and control of processes that transform inputs into services and products for internal, as well as external customers.

This book deals with managing those fundamental activities and processes that organizations use to produce goods and services that people use every day. A **process** is any activity or group of activities that takes one or more inputs, transforms them, and provides one or more outputs for its customers. For organizational purposes, processes tend to be clustered together into operations. An **operation** is a group of resources performing all or part of one or more processes. Processes can be linked together to form a **supply chain**, which is the interrelated series of processes within a firm and across different firms that produce a service or product to the satisfaction of customers.[1] A firm can have multiple supply chains, which vary by the product or service provided. **Supply chain management** is the synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services, and information with customer demand. For example, Scholastic must schedule the printing of a very large quantity of books in a timely fashion, receive orders from its largest customers, directly load and dispatch a fleet of trucks by specific destination while bypassing regular warehouses, keep track of their progress using technology, and finally, bill their customers and collect payment. The operational planning at Scholastic, along with internal and external coordination within its supply chain, provides one example of designing customized processes for competitive operations.

Operations and Supply Chain Management across the Organization

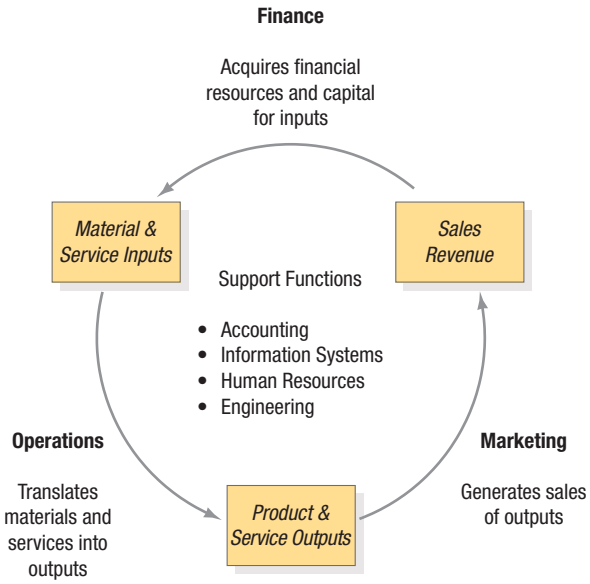
Broadly speaking, operations and supply chain management underlie all departments and functions in a business. Whether you aspire to manage a department or a particular process within it, or you just want to understand how the process you are a part of fits into the overall fabric of the business, you need to understand the principles of operations and supply chain management.

Operations serve as an excellent career path to upper management positions in many organizations. The reason is that operations managers are responsible for key decisions that affect the success of the organization. In manufacturing firms, the head of operations usually holds the

[1]The terms *supply chain* and *value chain* are sometimes used interchangeably.

title chief operations officer (COO) or vice president of manufacturing (or of production or operations). The corresponding title in a service organization might be COO or vice president (or director) of operations. Reporting to the head of operations are the managers of departments, such as customer service, production and inventory control, and quality assurance.

Figure 1.1 shows operations as one of the key functions within an organization. The circular relationships in Figure 1.1 highlight the importance of the coordination among the three mainline functions of any business, namely, (1) operations, (2) marketing, and (3) finance. Each function is unique and has its own knowledge and skill areas, primary responsibilities, processes, and decision domains. From an external perspective, finance generates resources, capital, and funds from investors and sales of its goods and services in the marketplace. Based on business strategy, the finance and operations functions then decide how to invest these resources and convert them into physical assets and material inputs. Operations subsequently transforms these material and service inputs into product and service outputs. These outputs must match the characteristics that can be sold in the selected markets by marketing. Marketing is responsible for producing sales revenue of the outputs, which become returns to investors and capital for supporting operations. Functions such as accounting, information systems, human resources, and engineering make the firm complete by providing essential information, services, and other managerial support.



▲ **FIGURE 1.1**
Integration between Different Functional Areas of a Business

These relationships provide direction for the business as a whole, and are aligned to the same strategic intent. It is important to understand the entire circle, and not just the individual functional areas. How well these functions work together determines the effectiveness of the organization. Functions should be integrated and should pursue a common strategy. Success depends on how well they are able to do so. No part of this circle can be dismissed or minimized without loss of effectiveness, and regardless of how departments and functions are individually managed, they are always linked together through processes. Thus, a firm competes not only by offering new services and products, creative marketing, and skillful finance, but also through its unique competencies in operations and sound management of core processes.

Historical Evolution of Operations and Supply Chain Management

The history of modern operations and supply chain management is rich and over two hundred years old, even though its practice has been around in one form or another for centuries. James Watt invented the steam engine in 1785. The subsequent establishment of railroads facilitated efficient movement of goods throughout Europe, and eventually even in distant colonies such as India. With the invention of the cotton gin in 1794, Eli Whitney introduced the concept of interchangeable parts. It revolutionized the art of machine-based manufacturing, and coupled with the invention of the steam engine, lead to the great industrial revolution in England and the rest of Europe. The textile industry was one of the earliest industries to be mechanized. The industrial revolution gradually spread to the United States and the rest of the world in the nineteenth century, and was accompanied by such great innovations as the internal combustion engine, steam-powered ships, metallurgy of iron making, large-scale production of chemicals, and invention of machine tools, among others. The foundations of modern manufacturing and technological breakthroughs were also inspired by the creation of a mechanical computer by Charles Babbage in the early part of the nineteenth century. He also pioneered the concept of division of labor, which laid the foundation for scientific management of operations and supply chain management that was further improved upon by Frederick Taylor in 1911.



Henry Ford with a Model T in Buffalo, New York, in 1921. The Ford Motor Company, founded in 1903, produced about one million Model T's in 1921.

AFP/Newscom

Three other landmark events from the twentieth century define the history of operations and supply chain management. First is the invention of the assembly line for the Model T car by Henry Ford in 1909. The era of mass production was born, where complex products like automobiles could be manufactured in large numbers at affordable prices through repetitive manufacturing. Second, Alfred Sloan in the 1930s introduced the idea of strategic planning for achieving product proliferation and variety, with the newly founded General Motors Corporation offering "a car for every purse and purpose." Finally, with the publication of the Toyota Production System in 1978, Taiichi Ohno laid the groundwork for removing wasteful activities from an organization, a concept that we explore further in this book while learning about lean systems.

The recent history of operations and supply chains over the past three decades has been steeped in technological advances. The 1980s were characterized by wide availability of computer aided design (CAD), computer aided manufacturing (CAM), and automation. Information technology applications started playing an increasingly important role in 1990s, and started connecting the firm with its extended enterprise through Enterprise Resource Planning Systems and outsourced technology hosting for supply chain solutions. Service organizations like Federal Express, United Parcel Service (UPS), and Walmart also became sophisticated users of information technology in operations, logistics, and management of supply chains. The new millennium has seen an acceleration of this trend, along with an increased focus on sustainability and the natural environment. We cover all these ideas and topical areas in greater detail throughout this book.

external customers

A customer who is either an end user or an intermediary (e.g., manufacturers, financial institutions, or retailers) buying the firm's finished services or products.

internal customers

One or more employees or processes that rely on inputs from other employees or processes in order to perform their work.

external suppliers

The businesses or individuals who provide the resources, services, products, and materials for the firm's short-term and long-term needs.

internal suppliers

The employees or processes that supply important information or materials to a firm's processes.

A Process View

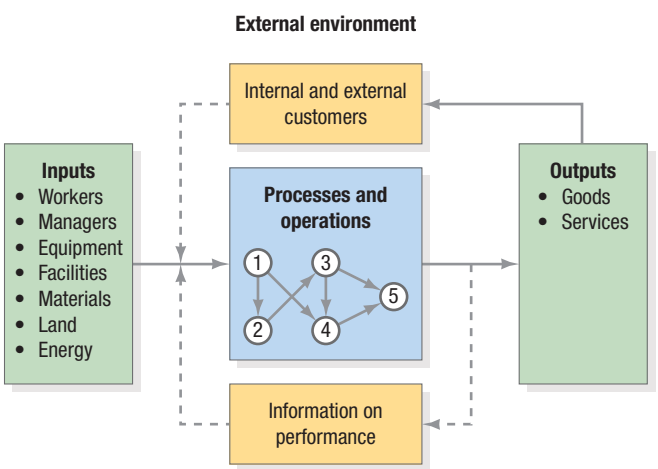
You might wonder why we begin by looking at processes, rather than at departments or even the firm. The reason is that a process view of the firm provides a much more relevant picture of the way firms actually work. Departments typically have their own set of objectives, a set of resources with capabilities to achieve those objectives, and managers and employees responsible for performance. Some processes, such as billing, may be so specific that they are contained wholly within a single department, such as accounting.

The concept of a process, however, can be much broader. A process can have its own set of objectives, involve a work flow that cuts across departmental boundaries, and require resources from several departments. You will see examples throughout this text of companies that discovered how to use their processes to gain a competitive advantage. You will notice that the key to success in many organizations is a keen understanding of how their processes work, since an organization is only as effective as its processes. Therefore, operations management is relevant and important for all students, regardless of major, because all departments have processes that must be managed effectively to gain a competitive advantage.

How Processes Work

Figure 1.2 shows how processes work in an organization. Any process has inputs and outputs. Inputs can include a combination of human resources (workers and managers), capital (equipment and facilities), purchased materials and services, land, and energy. The numbered circles in Figure 1.2 represent operations through which services, products, or customers pass and where processes are performed. The arrows represent flows, and can cross because one job or customer can have different requirements (and thus a different flow pattern) than the next job or customer.

▼ **FIGURE 1.2**
Processes and Operations



Processes provide outputs to customers. These outputs may often be services (that can take the form of information) or tangible products. Every process and every person in an organization has customers. Some are **external customers**, who may be end users or intermediaries (e.g., manufacturers, financial institutions, or retailers) buying the firm's finished services or products. Others are **internal customers**, who may be employees in the firm whose process inputs are actually the outputs of earlier processes managed within the firm. Either way, processes must be managed with the customer in mind.

In a similar fashion, every process and every person in an organization relies on suppliers. **External suppliers** may be other businesses or individuals who provide the resources, services, products, and materials for the firm's short-term and long-term needs. Processes also have **internal suppliers**, who may be employees or processes that supply important information or materials.

Inputs and outputs vary depending on the service or product provided. For example, inputs at a jewelry store include merchandise, the store building, registers, the jeweler, and customers; outputs to external customers are services and sold merchandise. Inputs to a factory manufacturing blue jeans include denim, machines, the plant, workers, managers, and services provided by outside consultants; outputs are clothing and supporting services. The fundamental role of inputs, processes, and customer outputs holds true for processes at all organizations.

Figure 1.2 can represent a whole firm, a department, a small group, or even a single individual. Each one has inputs and uses processes at various operations to provide outputs. The dashed lines represent two special types of input: participation by customers and information on performance from both internal and external sources. Participation by customers occurs not only when they receive outputs, but also when they take an active part in the processes, such as when students participate in a class discussion. Information on performance includes internal reports on customer service or inventory levels and external information from market research, government reports, or telephone calls from suppliers. Managers need all types of information to manage processes most effectively.

Nested Processes

Processes can be broken down into subprocesses, which in turn can be broken down further into still more subprocesses. We refer to this concept of a process within a process as a **nested process**. It may be helpful to separate one part of a process from another for several reasons. One person or one department may be unable to perform all parts of the process, or different parts of the process may require different skills. Some parts of the process may be designed for routine work while other parts may be geared for customized work. The concept of nested processes is illustrated in greater detail in Chapter 4, "Analyzing Processes," where we reinforce the need to understand and improve activities within a business and each process's inputs and outputs.

nested process
The concept of a process within a process.

Service and Manufacturing Processes

Two major types of processes are (1) service and (2) manufacturing. Service processes pervade the business world and have a prominent place in our discussion of operations management. Manufacturing processes are also important; without them the products we enjoy as part of our daily lives would not exist. In addition, manufacturing gives rise to service opportunities.

Differences Why do we distinguish between service and manufacturing processes? The answer lies at the heart of the design of competitive processes. While Figure 1.3 shows several distinctions between service and manufacturing processes along a continuum, the two key differences that we discuss in detail are (1) the nature of their output and (2) the degree of customer contact. In general, manufacturing processes also have longer response times, are more capital intensive, and their quality can be measured more easily than those of service processes.

Manufacturing processes convert materials into goods that have a physical form we call products. For example, an assembly line produces a 350 Z sports car, and a tailor produces an outfit for the rack of an upscale clothing store. The transformation processes change the materials on one or more of the following dimensions:

1. Physical properties
2. Shape
3. Size (e.g., length, breadth, and height of a rectangular block of wood)
4. Surface finish
5. Joining parts and materials



- Physical, durable output
- Output can be inventoried
- Low customer contact
- Long response time
- Capital intensive
- Quality easily measured

- Intangible, perishable output
- Output cannot be inventoried
- High customer contact
- Short response time
- Labor intensive
- Quality not easily measured

▲ **FIGURE 1.3**
Continuum of Characteristics of Manufacturing and Service Processes

The outputs from manufacturing processes can be produced, stored, and transported in anticipation of future demand.

If a process does not change the properties of materials on at least one of these five dimensions, it is considered a service (or nonmanufacturing) process. Service processes tend to produce intangible, perishable outputs. For example, the output from the auto loan process of a bank would be a car loan, and an output of the order fulfillment process of the U.S. Postal Service is the delivery of your letter. The outputs of service processes typically cannot be held in a finished goods inventory to insulate the process from erratic customer demands.

A second key difference between service processes and manufacturing processes is degree of customer contact. Service processes tend to have a higher degree of customer contact. Customers may take an active role in the process itself, as in the case of shopping in a supermarket, or they may be in close contact with the service provider to communicate specific needs, as in the case of a medical clinic. Manufacturing processes tend to have less customer contact. For example, washing machines are ultimately produced to meet retail forecasts. The process requires little information from the ultimate consumers (you and me), except indirectly through market surveys and market focus groups. Even though the distinction between service and manufacturing processes on the basis of customer contact is not perfect, the important point is that managers must recognize the degree of customer contact required when designing processes.

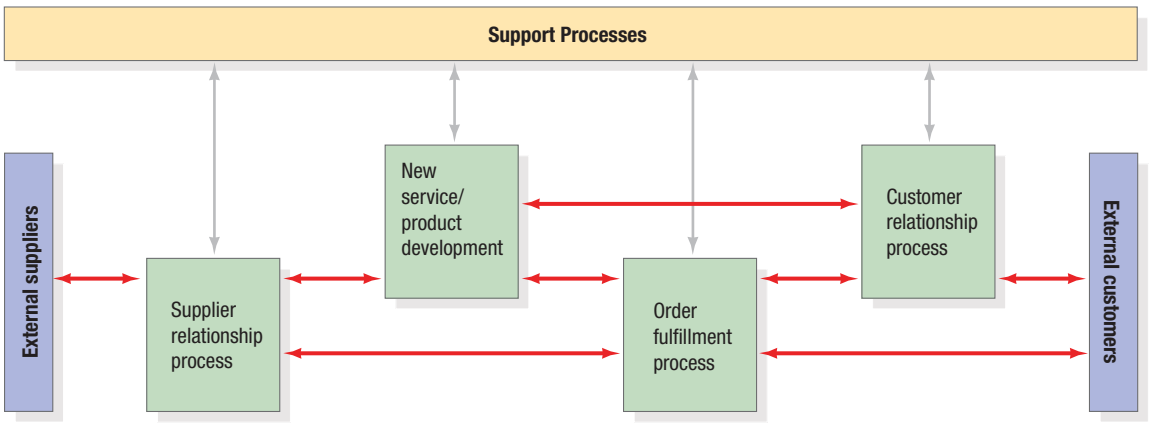
Similarities At the level of the firm, service providers do not just offer services and manufacturers do not just offer products. Patrons of a restaurant expect good service and good food. A customer purchasing a new computer expects a good product as well as a good warranty, maintenance, replacement, and financial services.

Further, even though service processes do not keep finished goods inventories, they do inventory their inputs. For example, hospitals keep inventories of medical supplies and materials needed for day-to-day operations. Some manufacturing processes, on the other hand, do not inventory their outputs because they are too costly. Such would be the case with low-volume customized products (e.g., tailored suits) or products with short shelf lives (e.g., daily newspapers).

When you look at what is being done at the process level, it is much easier to see whether the *process* is providing a service or manufacturing a product. However, this clarity is lost when the whole company is classified as either a manufacturer or a service provider because it often performs both types of processes. For example, the process of cooking a hamburger at a McDonald's is a manufacturing process because it changes the material's physical properties (dimension 1), as is the process of assembling the hamburger with the bun (dimension 5). However, most of the other processes visible or invisible to McDonald's customers are service processes. You can debate whether to call the whole McDonald's organization a service provider or a manufacturer, whereas classifications at the process level are much less ambiguous.

The Supply Chain View

Most services or products are produced through a series of interrelated business activities. Each activity in a process should add value to the preceding activities; waste and unnecessary cost should be eliminated. Our process view of a firm is helpful for understanding how services or products are produced and why cross-functional coordination is important, but it does not shed any light on the strategic benefits of the processes. The missing strategic insight is that processes must add value for customers throughout the supply chain. The concept of supply chains reinforces the link between processes and performance, which includes a firm's internal processes as well as those of its external customers and suppliers. It also focuses attention on the two main types of processes in the supply chain, namely (1) core processes and (2) support processes. Figure 1.4 shows the links between the core and support processes in a firm and a firm's external customers and suppliers within its supply chain.



▲ **FIGURE 1.4**
Supply Chain Linkages Showing Work and Information Flows

Core Processes

A **core process** is a set of activities that delivers value to external customers. Managers of these processes and their employees interact with external customers and build relationships with them, develop new services and products, interact with external suppliers, and produce the service or product for the external customer. Examples include a hotel's reservation handling, a new car design for an auto manufacturer, or Web-based purchasing for an online retailer like amazon.com. Of course, each of the core processes has nested processes within it.

In this text we focus on four core processes:

1. *Supplier Relationship Process.* Employees in the **supplier relationship process** select the suppliers of services, materials, and information and facilitate the timely and efficient flow of these items into the firm. Working effectively with suppliers can add significant value to the services or products of the firm. For example, negotiating fair prices, scheduling on-time deliveries, and gaining ideas and insights from critical suppliers are just a few of the ways to create value.
2. *New Service/Product Development Process.* Employees in the **new service/product development process** design and develop new services or products. The services or products may be developed to external customer specifications or conceived from inputs received from the market in general.
3. *Order Fulfillment Process.* The **order fulfillment process** includes the activities required to produce and deliver the service or product to the external customer.
4. *Customer Relationship Process,* sometimes referred to as *customer relationship management*. Employees involved in the **customer relationship process** identify, attract, and build relationships with external customers, and facilitate the placement of orders by customers. Traditional functions, such as marketing and sales, may be a part of this process.

core process

A set of activities that delivers value to external customers.

supplier relationship process

A process that selects the suppliers of services, materials, and information and facilitates the timely and efficient flow of these items into the firm.

new service/product development process

A process that designs and develops new services or products from inputs received from external customer specifications or from the market in general through the customer relationship process.

order fulfillment process

A process that includes the activities required to produce and deliver the service or product to the external customer.

customer relationship process

A process that identifies, attracts, and builds relationships with external customers, and facilitates the placement of orders by customers, sometimes referred to as *customer relationship management*.

Support Processes

A **support process** provides vital resources and inputs to the core processes and is essential to the management of the business. Firms have many support processes. Examples include budgeting, recruiting, and scheduling. Support processes provide key resources, capabilities, or other inputs that allow the core processes to function.

The Human Resources function in an organization provides many support processes such as recruiting and hiring workers who are needed at different levels of the organization, training the workers for skills and knowledge needed to properly execute their assigned responsibilities, and establishing incentive and compensation plans that reward employees for their performance. The legal department puts in place support processes that ensure that the firm is in compliance with the rules and regulations under which the business operates. The Accounting function supports processes that track how the firm's financial resources are being created and allocated over time, while the Information Systems function is responsible for the movement and processing of data and information needed to make business decisions. Support processes from different functional areas like Accounting, Engineering, Human Resources, and Information Systems are therefore vital to the execution of core processes highlighted in Figure 1.4.

support process

A process that provides vital resources and inputs to the core processes and therefore is essential to the management of the business.

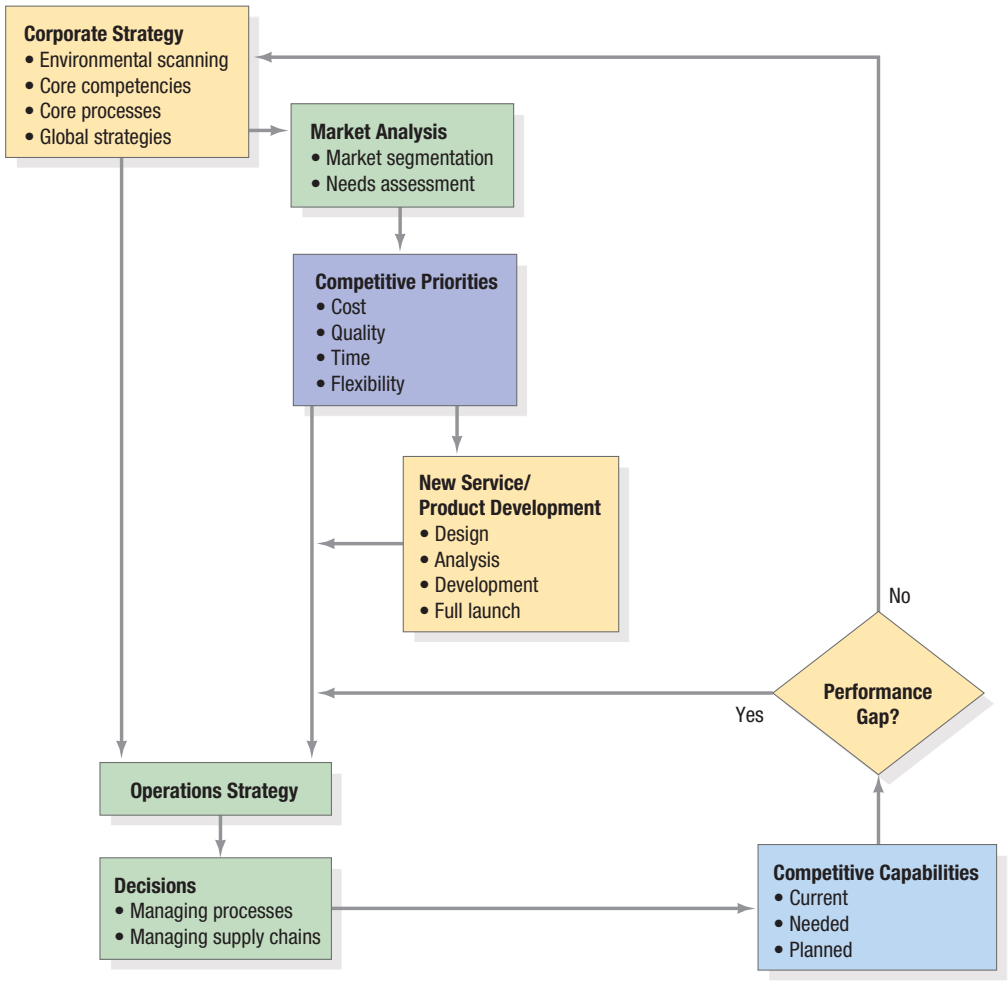
Operations Strategy

Operations strategy specifies the means by which operations implements corporate strategy and helps to build a customer-driven firm. It links long-term and short-term operations decisions to corporate strategy and develops the capabilities the firm needs to be competitive. It is at the heart of managing processes and supply chains. A firm's internal processes are only building blocks: They need to be organized to ultimately be effective in a competitive environment. Operations strategy is the linchpin that brings these processes together to form supply chains that extend beyond the walls of the firm, encompassing suppliers as well as customers. Since customers constantly desire change, the firm's operations strategy must be driven by the needs of its customers.

operations strategy

The means by which operations implements the firm's corporate strategy and helps to build a customer-driven firm.

Developing a customer-driven operations strategy begins with *corporate strategy*, which, as shown in Figure 1.5, coordinates the firm's overall goals with its core processes. It determines the markets the firm will serve and the responses the firm will make to changes in the environment. It provides the resources to develop the firm's core competencies and core processes, and it identifies the strategy the firm will employ in international markets. Based on corporate strategy, a *market analysis* categorizes the firm's customers, identifies their needs, and assesses competitors' strengths. This information is used to develop *competitive priorities*. These priorities help managers develop the services or products and the processes needed to be competitive in the



▶ **FIGURE 1.5**
Connection Between Corporate Strategy and Key Operations Management Decisions

marketplace. Competitive priorities are important to the design of existing as well as new services or products, the processes that will deliver them, and the operations strategy that will develop the firm's capabilities to fulfill them. Developing a firm's operations strategy is a continuous process because the firm's capabilities to meet the competitive priorities must be periodically checked and any gaps in performance must be addressed in the operations strategy.

Corporate Strategy

Corporate strategy provides an overall direction that serves as the framework for carrying out all the organization's functions. It specifies the business or businesses the company will pursue, isolates new opportunities and threats in the environment, and identifies growth objectives.

Developing a corporate strategy involves four considerations: (1) monitoring and adjusting to changes in the business environment, (2) identifying and developing the firm's core competencies, (3) developing the firm's core processes, and (4) developing the firm's global strategies.

Environmental Scanning The external business environment in which a firm competes changes continually and an organization needs to adapt to those changes. Adaptation begins with *environmental scanning*, the process by which managers monitor trends in the environment (e.g., the industry, the marketplace, and society) for potential opportunities or threats. A crucial reason for environmental scanning is to stay ahead of the competition. Competitors may be gaining an edge by broadening service or product lines, improving quality, or lowering costs. New entrants into the market or competitors that offer substitutes for a firm's service or product may threaten continued profitability. Other important environmental concerns include economic trends, technological changes, political conditions, social changes (i.e., attitudes toward work), and the availability of vital resources. For example, car manufacturers recognize that dwindling oil reserves will eventually require alternative fuels for their cars. Consequently, they have designed prototype cars that use hydrogen or electric power as supplements to gasoline as a fuel.

Developing Core Competencies Good managerial skill alone cannot overcome environmental changes. Firms succeed by taking advantage of what they do particularly well—that is, the organization's unique strengths. **Core competencies** are the unique resources and strengths that an organization's management considers when formulating strategy. They reflect the collective learning of the organization, especially in how to coordinate processes and integrate technologies. These competencies include the following:

core competencies
The unique resources and strengths that an organization's management considers when formulating strategy.

1. *Workforce.* A well-trained and flexible workforce allows organizations to respond to market needs in a timely fashion. This competency is particularly important in service organizations, where customers come in direct contact with employees.
2. *Facilities.* Having well-located facilities (offices, stores, and plants) is a primary advantage because of the long **lead time** needed to build new ones. In addition, flexible facilities that can handle a variety of services or products at different levels of volume provide a competitive advantage.
3. *Market and Financial Know-How.* An organization that can easily attract capital from stock sales, market and distribute its services or products, or differentiate them from similar services or products on the market has a competitive edge.
4. *Systems and Technology.* Organizations with expertise in information systems have an edge in industries that are data intensive, such as banking. Particularly advantageous is expertise in Internet technologies and applications, such as business-to-consumer and business-to-business systems. Having the patents on a new technology is also a big advantage.

lead time
The elapsed time between the receipt of a customer order and filling it.

Developing Core Processes A firm's core competencies should drive its core processes: customer relationship, new service/product development, order fulfillment, and supplier relationship. Many companies have all four processes, while others focus on a subset of them to better match their core competencies, since they find it difficult to be good at all four processes and still be competitive. For instance, in the credit card business within the banking industry, some companies primarily specialize in finding customers and maintaining relationships with them. American Airlines's credit card program reaches out and achieves a special affinity to customers through its marketing database. On the other hand, specialized credit card companies, such as CapitalOne, focus on service innovation by creating new features and pricing programs. Finally, many companies are taking over the order fulfillment process by managing the processing of credit card transactions and call centers. The important point is that every firm must evaluate its core competencies and choose to focus on those processes that provide it the greatest competitive strength.



Romeo Gacad/AFP/Getty Images/Newscom

The popular smiling red bee, the mascot of Jollibee, welcomes customers at an outlet in Manila. What began from a two ice cream parlors in Manila in 1975, Jollibee has grown into the biggest Philippines fast-food company employing over 26,000 people in over 1,000 stores in seven countries. By catering to local tastes and preferences, Jollibee took 65 percent of the fiercely competitive Philippine fast-food market, pushing the world giant McDonald's into second place.

Global Strategies Identifying opportunities and threats today requires a global perspective. A global strategy may include buying foreign services or parts, combating threats from foreign competitors, or planning ways to enter markets beyond traditional national boundaries. Although warding off threats from global competitors is necessary, firms should also actively seek to penetrate foreign markets. Two effective global strategies are (1) strategic alliances and (2) locating abroad.

One way for a firm to open foreign markets is to create a *strategic alliance.* A strategic alliance is an agreement with another firm that may take one of three forms. One form of strategic alliance is the *collaborative effort,* which often arises when one firm has core competencies that another needs but is unwilling (or unable) to duplicate. Such arrangements commonly arise out of buyer–supplier relationships.

Another form of strategic alliance is the *joint venture,* in which two firms agree to produce a service or product jointly. This approach is often used by firms to gain access to foreign markets. For example, to get access to the large Chinese market, General Motors (GM) and Volkswagen (VW) each developed joint ventures with Shanghai Automotive Industry Corporation or SAIC.[2] The Chinese partner is a large manufacturer of automobiles, producing more than 600,000 cars with GM and VW. In 2010, SAIC upped its total share to 51% in Shanghai GM, which is now among the top three passenger vehicle producers in mainland China. Finally, *technology licensing* is a form of strategic alliance in which one company licenses its service or production methods to another. Licenses may be used to gain access to foreign markets.

Another way to enter global markets is to locate operations in a foreign country. However, managers must recognize that what works well in their home country might not work well elsewhere. The economic and political environment or customers' needs may be significantly different. For example, the family-owned chain Jollibee Foods Corporation has become the dominant fast-food chain in the Philippines by catering to a local preference for sweet and spicy flavors, which it incorporates into its fried chicken, spaghetti, and burgers. Jollibee's strength is its creative marketing programs and an understanding of local tastes and claims that its burger is similar to the one a Filipino would cook at home. McDonald's responded by introducing its own Filipino-style spicy burger, but competition is stiff. This example shows that to be successful, corporate strategies must recognize customs, preferences, and economic conditions in other countries.

Locating abroad is a key decision in the design of supply chains because it affects the flow of materials, information, and employees in support of the firm's core processes. Chapter 10, "Designing Effective Supply Chains," and Chapter 11, "Locating Facilities," offer more in-depth discussion of these other implications.

Market Analysis

One key to successfully formulating a customer-driven operations strategy for both service and manufacturing firms is to understand what the customer wants and how to provide it. A *market analysis* first divides the firm's customers into market segments and then identifies the needs of each segment. In this section, we examine the process of market analysis and we define and discuss the concepts of market segmentation and needs assessment.

Market Segmentation *Market segmentation* is the process of identifying groups of customers with enough in common to warrant the design and provision of services or products that the group wants and needs. To identify market segments, the analyst must determine the characteristics that clearly differentiate each segment. The company can then develop a sound marketing program and an effective operating strategy to support it. For instance, The Gap, Inc., a major provider of casual clothes, targets teenagers and young adults while the parents or guardians of infants through 12-year-olds are the primary targets for its GapKids stores. At one time, managers thought of customers as a homogeneous mass market, but now realize that two customers may use the same product for different reasons. Identifying the key factors in each market segment is the starting point in devising a customer-driven operations strategy.

Needs Assessment The second step in market analysis is to make a *needs assessment*, which identifies the needs of each segment and assesses how well competitors are addressing those needs. Each market segment's needs can be related to the service or product and its supply chain. Market needs should include both the tangible and intangible attributes and features of products and services that a customer desires. Market needs may be grouped as follows:

- *Service or Product Needs.* Attributes of the service or product, such as price, quality, and degree of customization.
- *Delivery System Needs.* Attributes of the processes and the supporting systems, and resources needed to deliver the service or product, such as availability, convenience, courtesy, safety, accuracy, reliability, delivery speed, and delivery dependability.
- *Volume Needs.* Attributes of the demand for the service or product, such as high or low volume, degree of variability in volume, and degree of predictability in volume.
- *Other Needs.* Other attributes, such as reputation and number of years in business, after-sale technical support, ability to invest in international financial markets, and competent legal services.

[2]Alex Taylor, "Shanghai Auto Wants to Be the World's Next Great Car Company," *Fortune* (October 4, 2004), pp. 103–110.

Once it makes this assessment, the firm can incorporate the needs of customers into the design of the service or product and the supply chain that must deliver it. We further discuss these new service and product development-related issues in Chapter 12, "Integrating the Supply Chain."

Competitive Priorities and Capabilities

A customer-driven operations strategy requires a cross-functional effort by all areas of the firm to understand the needs of the firm's external customers and to specify the operating capabilities the firm requires to outperform its competitors. Such a strategy also addresses the needs of internal customers because the overall performance of the firm depends upon the performance of its core and supporting processes, which must be coordinated to provide the overall desirable outcome for the external customer.

Competitive priorities are the critical operational dimensions a process or supply chain must possess to satisfy internal or external customers, both now and in the future. Competitive priorities are planned for processes and the supply chain created from them. They must be present to maintain or build market share or to allow other internal processes to be successful. Not all competitive priorities are critical for a given process; management selects those that are most important. **Competitive capabilities** are the cost, quality, time, and flexibility dimensions that a process or supply chain actually possesses and is able to deliver. When the capability falls short of the priority attached to it, management must find ways to close the gap or else revise the priority.

competitive priorities
The critical dimensions that a process or supply chain must possess to satisfy its internal or external customers, both now and in the future.

competitive capabilities
The cost, quality, time, and flexibility dimensions that a process or supply chain actually possesses and is able to deliver.

We focus on nine broad competitive priorities that fall into the four capability groups of cost, quality, time, and flexibility. Table 1.1 provides definitions and examples of these competitive priorities, as well as how firms achieve them at the process level.

At times, management may emphasize a cluster of competitive priorities together. For example, many companies focus on the competitive priorities of delivery speed and development speed for their processes, a strategy called **time-based competition**. To implement the strategy, managers carefully define the steps and time needed to deliver a service or produce a product and then critically analyze each step to determine whether they can save time without hurting quality.

time-based competition
A strategy that focuses on the competitive priorities of delivery speed and development speed.

To link to corporate strategy, management assigns selected competitive priorities to each process (and the supply chains created from them) that are consistent with the needs of external as well as internal customers. Competitive priorities may change over time. For example, consider a high-volume standardized product, such as color ink-jet desktop printers. In the early stages of the ramp-up period when the printers had just entered the mass market, the manufacturing processes required consistent quality, delivery speed, and volume flexibility. In the later stages of the ramp-up when demand was high, the competitive priorities became low-cost operations, consistent quality, and on-time delivery. Competitive priorities must change and evolve over time along with changing business conditions and customer preferences.



The lavish interior lobby decor of the Ritz Carlton resort in Palm Beach, Florida, USA

America/Alamy

Order Winners and Qualifiers

Competitive priorities focus on what operations can do to help a firm be more competitive, and are in response to what the market wants. Another useful way to examine a firm's ability to be successful in the marketplace is to identify the order winners and order qualifiers. An **order winner** is a criterion that customers use to differentiate the services or products of one firm from those of another. Order winners can include price (which is supported by low-cost operations) and other dimensions of quality, time, and flexibility. However, order winners also include criteria not directly related to the firm's operations, such as after-sale support (Are maintenance service contracts available? Is there a return policy?); technical support (What help do I get if something goes wrong? How knowledgeable are the technicians?); and reputation (How long has this company been in business? Have other customers been satisfied with the service or product?). It may take good performance on a subset of the order-winner criteria, cutting across operational as well as nonoperational criteria, to make a sale.

order winner
A criterion customers use to differentiate the services or products of one firm from those of another.

Order winners are derived from the considerations customers use when deciding which firm to purchase a service or product from in a given market segment. Sometimes customers demand a certain level of demonstrated performance before even contemplating a service or product.

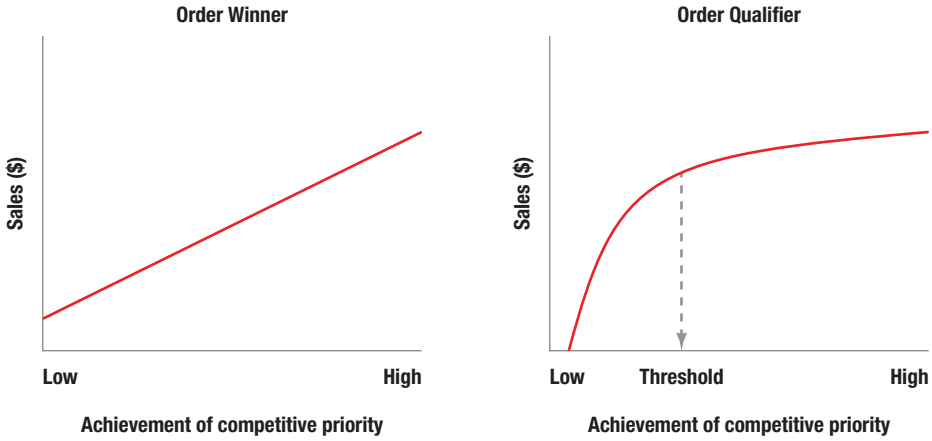
TABLE 1.1 | DEFINITIONS, PROCESS CONSIDERATIONS, AND EXAMPLES OF COMPETITIVE PRIORITIES

Cost	Definition	Processes Considerations	Example
1. **Low-cost operations**	Delivering a service or a product at the lowest possible cost to the satisfaction of external or internal customers of the process or supply chain	To reduce costs, processes must be designed and operated to make them efficient using rigorous process analysis that addresses workforce, methods, scrap or rework, overhead, and other factors, such as investments in new automated facilities or technologies to lower the cost per unit of the service or product.	**Costco** achieves low costs by designing all processes for efficiency, stacking products on pallets in warehouse-type stores, and negotiating aggressively with their suppliers. Costco can provide low prices to its customers because they have designed operations for low cost.
Quality			
2. **Top quality**	Delivering an outstanding service or product	To deliver top quality, a service process may require a high level of customer contact, and high levels of helpfulness, courtesy, and availability of servers. It may require superior product features, close tolerances, and greater durability from a manufacturing process.	**Rolex** is known globally for creating precision timepieces.
3. **Consistent quality**	Producing services or products that meet design specifications on a consistent basis	Processes must be designed and monitored to reduce errors, prevent defects, and achieve similar outcomes over time, regardless of the "level" of quality.	**McDonald's** standardizes work methods, staff training processes, and procurement of raw materials to achieve the same consistent product and process quality from one store to the next.
Time			
4. **Delivery speed**	Quickly filling a customer's order	Design processes to reduce lead time (elapsed time between the receipt of a customer order and filling it) through keeping backup capacity cushions, storing inventory, and using premier transportation options.	**Dell** engineered its customer relationship, order fulfillment, and supplier relationship processes to create an integrated and an agile supply chain that delivers reliable and inexpensive computers to its customers with short lead times.
5. **On-time delivery**	Meeting delivery-time promises	Along with processes that reduce lead time, planning processes (forecasting, appointments, order promising, scheduling, and capacity planning) are used to increase percent of customer orders shipped when promised (95% is often a typical goal).	**United Parcel Services (UPS)** uses its expertise in logistics and warehousing processes to deliver a very large volume of shipments on-time across the globe.
6. **Development speed**	Quickly introducing a new service or a product	Processes aim to achieve cross-functional integration and involvement of critical external suppliers in the service or product development process.	**Zara** is known for its ability to bring fashionable clothing designs from the runway to market quickly.
Flexibility			
7. **Customization**	Satisfying the unique needs of each customer by changing service or product designs	Processes with a customization strategy typically have low volume, close customer contact, and an ability to reconfigure processes to meet diverse types of customer needs.	**Ritz Carlton** customizes services to individual guest preferences.
8. **Variety**	Handling a wide assortment of services or products efficiently	Processes supporting variety must be capable of larger volumes than processes supporting customization. Services or products are not necessarily unique to specific customers and may have repetitive demands.	**Amazon.com** uses information technology and streamlined customer relationship and order fulfillment processes to reliably deliver a vast variety of items to its customers.
9. **Volume flexibility**	Accelerating or decelerating the rate of production of services or products quickly to handle large fluctuations in demand	Processes must be designed for excess capacity and excess inventory to handle demand fluctuations that can vary in cycles from days to months. This priority could also be met with a strategy that adjusts capacity without accumulation of inventory or excess capacity.	**The United States Post Office (USPS)** can have severe demand peak fluctuations at large postal facilities where processes are flexibly designed for receiving, sorting, and dispatching mail to numerous branch locations.

Minimal level required from a set of criteria for a firm to do business in a particular market segment is called an **order qualifier**. Fulfilling the order qualifier will not ensure competitive success; it will only position the firm to compete in the market. From an operations perspective, understanding which competitive priorities are order qualifiers and which ones are order winners is important for the investments made in the design and management of processes and supply chains.

order qualifier
Minimal level required from a set of criteria for a firm to do business in a particular market segment.

Figure 1.6 shows how order winners and qualifiers are related to achieving the competitive priorities of a firm. If a minimum threshold level is not met for an order-qualifying dimension (consistent quality, for example) by a firm, then it would get disqualified from even being considered further by its customers. For example, there is a level of quality consistency that is minimally tolerable by customers in the auto industry. When the subcompact car Yugo built by Zastava Corporation could not sustain the minimal level of quality, consistency, and reliability expected by customers, it had to exit the U.S. car market in 1991 despite offering very low prices (order winner) of under $4,000. However, once the firm qualifies by attaining consistent quality beyond the threshold, it may only gain additional sales at a very low rate by investing further in improving that order-qualifying dimension. In contrast, for an order-winning dimension (i.e., low price driven by low-cost operations), a firm can reasonably expect to gain appreciably greater sales and market share by continuously lowering its prices as long as the order qualifier (i.e., consistent quality) is being adequately met. Toyota Corolla and Honda Civic have successfully followed this route in the marketplace to become leaders in their target market segment.



▲ **FIGURE 1.6**
Relationship of Order Winners and Order Qualifiers to Competitive Priorities

Order winners and qualifiers are often used in competitive bidding. For example, before a buyer considers a bid, suppliers may be required to document their ability to provide consistent quality as measured by adherence to the design specifications for the service or component they are supplying (order qualifier). Once qualified, the supplier may eventually be selected by the buyer on the basis of low prices (order winner) and the reputation of the supplier (order winner).

Using Competitive Priorities: An Airline Example

To get a better understanding of how companies use competitive priorities, let us look at a major airline. We will consider two market segments: (1) first-class passengers and (2) coach passengers. Core services for both market segments are ticketing and seat selection, baggage handling, and transportation to the customer's destination. The peripheral services are quite different across the two market segments. First-class passengers require separate airport lounges; preferred treatment during check-in, boarding, and deplaning; more comfortable seats; better meals and beverages; more personal attention (cabin attendants who refer to customers by name); more frequent service from attendants; high levels of courtesy; and low volumes of passengers (adding to the feeling of being special). Coach passengers are satisfied with standardized services (no surprises), courteous flight attendants, and low prices. Both market segments expect the airline to hold to its schedule. Consequently, we can say that the competitive priorities for the



One of the competitive priorities of airline companies is on-time delivery of their services. Being able to repair and maintain planes rapidly to avoid delays is a crucial aspect of this.

Newscom

first-class segment are *top quality* and *on-time delivery*, whereas the competitive priorities for the coach segment are *low-cost operations*, *consistent quality*, and *on-time delivery*.

The airline knows what its collective capabilities must be as a firm, but how does that get communicated to each of its core processes? Let us focus on the four core processes: (1) customer relationship, (2) new service/product development, (3) order fulfillment, and (4) supplier relationship. Competitive priorities are assigned to each core process to achieve the service required to provide complete customer satisfaction. Table 1.2 shows some possible assignments, just to give you an idea of how this works.

TABLE 1.2 | COMPETITIVE PRIORITIES ACROSS DIFFERENT CORE PROCESSES FOR AN AIRLINE

	CORE PROCESSES			
Priority	**Supplier Relationship**	**New Service Development**	**Order Fulfillment**	**Customer Relationship**
Low Cost Operations	Costs of acquiring inputs must be kept to a minimum to allow for competitive pricing.		Airlines compete on price and must keep operating costs in check.	
Top Quality		New services must be carefully designed because the future of the airline industry depends on them.	High quality meal and beverage service delivered by experienced cabin attendants ensures that the service provided to first-class passengers is kept top notch.	High levels of customer contact and lounge service for the first-class passengers.
Consistent Quality	Quality of the inputs must adhere to the required specifications. In addition, information provided to suppliers must be accurate.		Once the quality level is set, it is important to achieve it every time.	The information and service must be error free.
Delivery Speed				Customers want immediate information regarding flight schedules and other ticketing information.
On time delivery	Inputs must be delivered to tight schedules.		The airline strives to arrive at destinations on schedule, otherwise passengers might miss connections to other flights.	
Development Speed		It is important to get to the market fast to preempt the competition.		
Customization		The process must be able to create unique services.		
Variety	Many different inputs must be acquired, including maintenance items, meals and beverages.		Maintenance operations are required for a variety of aircraft models.	The process must be capable of handling the service needs of all market segments and promotional programs.
Volume Flexibility	The process must be able to handle variations in supply quantities efficiently.			

Operations Strategy as a Pattern of Decisions

Operations strategy translates service or product plans and competitive priorities for each market segment into decisions affecting the supply chains that support those market segments. Even if it is not formally stated, the current operations strategy for any firm is really the pattern of decisions that have been made for its processes and supply chains. As we have previously seen in Figure 1.5, corporate strategy provides the umbrella for key operations management decisions that contribute to the development of the firm's ability to compete successfully in the marketplace. Once managers determine the competitive priorities for a process, it is necessary to assess the *competitive capabilities* of the process. Any gap between a competitive priority and the capability to achieve that competitive priority must be closed by an effective operations strategy.

Developing capabilities and closing gaps is the thrust of operations strategy. To demonstrate how this works, suppose the management of a bank's credit card division decides to embark on a marketing campaign to significantly increase its business, while keeping costs low. A key process in this division is billing and payments. The division receives credit transactions from the merchants, pays the merchants, assembles and sends the bills to the credit card holders, and processes payments. The new marketing effort is expected to significantly increase the volume of bills and payments. In assessing the capabilities, the process must have to serve the bank's customers and to meet the challenges of the new market campaign; management assigns the following competitive priorities for the billing and payments process:

- *Low-Cost Operations.* It is important to maintain low costs in the processing of the bills because profit margins are tight.
- *Consistent Quality.* The process must consistently produce bills, make payments to the merchants, and record payments from the credit card holders accurately.
- *Delivery Speed.* Merchants want to be paid for the credit purchases quickly.
- *Volume Flexibility.* The marketing campaign is expected to generate many more transactions in a shorter period of time.

Management assumed that customers would avoid doing business with a bank that could not produce accurate bills or payments. Consequently, consistent quality is an order qualifier for this process.

Is the billing and payment process up to the competitive challenge? Table 1.3 shows how to match capabilities to priorities and uncover any gaps in the credit card division's operations strategy. The procedure for assessing an operations strategy begins with identifying good measures for each priority. The more quantitative the measures are, the better. Data are gathered for each measure to determine the current capabilities of the process. Gaps are identified by comparing each capability to management's target values for the measures, and unacceptable gaps are closed by appropriate actions.

The credit card division shows significant gaps in the process's capability for low-cost operations. Management's remedy is to redesign the process in ways that reduce costs but will not impair the other competitive priorities. Likewise, for volume flexibility, management realized that a high level of utilization is not conducive for processing quick surges in volumes while maintaining delivery speed. The recommended actions will help build a capability for meeting more volatile demands.

TABLE 1.3 | OPERATIONS STRATEGY ASSESSMENT OF THE BILLING AND PAYMENT PROCESS

Competitive Priority	Measure	Capability	Gap	Action
Low-cost operations	■ Cost per billing statement ■ Weekly postage	■ $0.0813 ■ $17,000	■ Target is $0.06 ■ Target is $14,000	■ Eliminate microfilming and storage of billing statements ■ Develop Web-based process for posting bills
Consistent quality	■ Percent errors in bill information ■ Percent errors in posting payments	■ 0.90% ■ 0.74%	■ Acceptable ■ Acceptable	■ No action ■ No action
Delivery speed	■ Lead time to process merchant payments	■ 48 hours	■ Acceptable	■ No action
Volume flexibility	■ Utilization	■ 98%	■ Too high to support rapid increase in volumes	■ Acquire temporary employees ■ Improve work methods

Trends in Operations Management

Several trends are currently having a great impact on operations management: productivity improvement; global competition; and ethical, workforce diversity, and environmental issues. Accelerating change in the form of information technology, e-commerce, robotics, and the Internet is dramatically affecting the design of new services and products as well as a firm's sales, order fulfillment, and purchasing processes. In this section, we look at these trends and their challenges for operations managers.

Productivity Improvement

productivity
The value of outputs (services and products) produced divided by the values of input resources (wages, costs of equipment, and so on).

Productivity is a basic measure of performance for economies, industries, firms, and processes. Improving productivity is a major trend in operations management because all firms face pressures to improve their processes and supply chains so as to compete with their domestic and foreign competitors. **Productivity** is the value of outputs (services and products) produced divided by the values of input resources (wages, cost of equipment, and so on) used:

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$

Manufacturing employment peaked at just below 20 million in mid-1979, and shrunk by nearly 8 million from 1979 to 2011.[3] However, the manufacturing productivity in the United States has climbed steadily, as more manufacturing capacity and output has been achieved efficiently with a leaner work force. It is interesting and even surprising to compare productivity improvements in the service and manufacturing sectors. In the United States, employment in the service sector has grown rapidly, outstripping the manufacturing sector. It now employs about 90 percent of the workforce. But service-sector productivity gains have been much lower. If productivity growth in the service sector stagnates, so does the overall standard of living regardless of which part of the world you live in. Other major industrial countries, such as Japan and Germany, are experiencing the same problem. Yet, signs of improvement are appearing. The surge of investment across national boundaries can stimulate productivity gains by exposing firms to greater competition. Increased investment in information technology by service providers also increases productivity.

Measuring Productivity As a manager, how do you measure the productivity of your processes? Many measures are available. For example, value of output can be measured by what the customer pays or simply by the number of units produced or customers served. The value of inputs can be judged by their cost or simply by the number of hours worked.

Managers usually pick several reasonable measures and monitor trends to spot areas needing improvement. For example, a manager at an insurance firm might measure office productivity as the number of insurance policies processed per employee per week. A manager at a carpet company might measure the productivity of installers as the number of square yards of carpet installed per hour. Both measures reflect *labor productivity*, which is an index of the output per person or per hour worked. Similar measures may be used for *machine productivity*, where the denominator is the number of machines. Accounting for several inputs simultaneously is also possible. *Multifactor productivity* is an index of the output provided by more than one of the resources used in production; it may be the value of the output divided by the sum of labor, materials, and overhead costs. Here is an example:

EXAMPLE 1.1 **Productivity Calculations**

MyOMLab
Tutor 1.1 in MyOMLab provides a new example for calculating productivity.

Calculate the productivity for the following operations:

a. Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week.

b. A team of workers makes 400 units of a product, which is sold in the market for $10 each. The accounting department reports that for this job the actual costs are $400 for labor, $1,000 for materials, and $300 for overhead.

[3] Paul Wiseman, "Despite China's Might, US Factories Maintain Edge," *The State* and *The Associated Press* (January 31, 2011).

SOLUTION

a. $$\text{Labor productivity} = \frac{\text{Policies processed}}{\text{Employee hours}} = \frac{600 \text{ policies}}{(3 \text{ employees})\,(40 \text{ hours/employee})} = 5 \text{ policies/hour}$$

b. $$\text{Multifactor productivity} = \frac{\text{Value of output}}{\text{Labor cost} + \text{Materials cost} + \text{Overhead cost}} = \frac{(400 \text{ units})\,(\$10/\text{unit})}{\$400 + \$1{,}000 + \$300} = \frac{\$4{,}000}{\$1{,}700} = 2.35$$

DECISION POINT

We want multifactor productivity to be as high as possible. These measures must be compared with performance levels in prior periods and with future goals. If they do not live up to expectations, the process should be investigated for improvement opportunities.

The Role of Management The way processes are managed plays a key role in productivity improvement. Managers must examine productivity from the level of the supply chain because it is the collective performance of individual processes that makes the difference. The challenge is to increase the value of output relative to the cost of input. If processes can generate more output or output of better quality using the same amount of input, productivity increases. If they can maintain the same level of output while reducing the use of resources, productivity also increases.

Global Competition

Most businesses realize that, to prosper, they must view customers, suppliers, facility locations, and competitors in global terms. Firms have found that they can increase their market penetration by locating their production facilities in foreign countries because it gives them a local presence that reduces customer aversion to buying imports. Globalization also allows firms to balance cash flows from other regions of the world when economic conditions are less robust in the home country. Sonoco, a $4-billion-a-year industrial and consumer packaging company in Hartsville, South Carolina, has 335 locations worldwide in Australia, China, Europe, Mexico, New Zealand, and Russia, with 41 industrial product manufacturing facilities and 6 paper mills in Europe alone. These global operations resulted in international sales and income growth even as domestic sales were stumbling during 2007. How did Sonoco do it?[4] Locating operations in countries with favorable tax laws is one reason. Lower tax rates in Italy and Canada helped in padding the earnings margin. Another reason was a weak dollar, whereby a $46 million boost came from turning foreign currencies into dollars as Sonoco exported such items as snack bag packaging, and tubes and cores used to hold tape and textiles, to operations it owned in foreign countries. The exchange



PR Newswire/Associated Press

Sonoco is a leading global manufacturer of industrial and consumer packaging goods with more than 300 locations in 35 countries serving 85 nations.

[4] Ben Werner, "Sonoco Holding Its Own," *The State* (February 7, 2008); **www.sonoco.com,** 2008.

rate difference was more than enough to counter the added expense of increased raw materials, shipping, and energy costs in the United States.

Most products today are composites of materials and services from all over the world. Your Gap polo shirt is sewn in Honduras from cloth cut in the United States. Sitting in a Cineplex theater (Canadian), you munch a Nestle's Crunch bar (Swiss) while watching a Columbia Pictures movie (Japanese). Five developments spurred the need for sound global strategies: (1) improved transportation and communications technologies, (2) loosened regulations on financial institutions, (3) increased demand for imported services and goods, (4) reduced import quotas and other international trade barriers due to the formation of regional trading blocks, such as the European Union (EU) and the North American Free Trade Agreement (NAFTA), and (5) comparative cost advantages.

Comparative Cost Advantages China and India have traditionally been the sources for low-cost, but skilled, labor, even though the cost advantage is diminishing as these countries become economically stronger. In the late 1990s, companies manufactured products in China to grab a foothold in a huge market, or to get cheap labor to produce low-tech products despite doubts about the quality of the workforce and poor roads and rail systems. Today, however, China's new factories, such as those in the Pudong industrial zone in Shanghai, produce a wide variety of products that are sold overseas in the United States and other regions of the world. U.S. manufacturers have increasingly abandoned low profit margin sectors like consumer electronics, shoes, and toys to emerging nations such as China and Indonesia. Instead, they are focusing on making expensive goods like computer chips, advanced machinery, and health care products that are complex and which require specialized labor.

Foreign companies have opened tens of thousands of new facilities in China over the past decade. Many goods the United States imports from China now come from foreign-owned companies with operations there. These companies include telephone makers, such as Nokia and Motorola, and nearly all of the big footwear and clothing brands. Many more major manufacturers are there as well. The implications for competition are enormous. Companies that do not have operations in China are finding it difficult to compete on the basis of low prices with companies that do. Instead, they must focus on speed and small production runs.

What China is to manufacturing, India is to service. As with the manufacturing companies, the cost of labor is a key factor. Indian software companies have grown sophisticated in their applications and offer a big advantage in cost. The computer services industry is also affected. Back-office operations are affected for the same reason. Many firms are using Indian companies for accounting and bookkeeping, preparing tax returns, and processing insurance claims. Many tech companies, such as Intel and Microsoft, are opening significant research and development (R&D) operations in India.



Yuriko Nakao/Reuters/Corbis

Shortage of components from suppliers prevented Nintendo from meeting the customer demand for its popular Wii game system.

Disadvantages of Globalization Of course, operations in other countries can have disadvantages. A firm may have to relinquish proprietary technology if it turns over some of its component manufacturing to offshore suppliers or if suppliers need the firm's technology to achieve desired quality and cost goals. Political risks may also be involved. Each nation can exercise its sovereignty over the people and property within its borders. The extreme case is nationalization, in which a government may take over a firm's assets without paying compensation. Exxon and other large multinational oil firms are scaling back operations in Venezuela due to nationalization concerns. Further, a firm may actually alienate customers back home if jobs are lost to offshore operations.

Employee skills may be lower in foreign countries, requiring additional training time. South Korean firms moved much of their sports shoe production to low-wage Indonesia and China, but they still manufacture hiking shoes and in-line roller skates in South Korea because of the greater

MANAGERIAL PRACTICE 1.1 Japanese Earthquake and its Supply Chain Impact

Northeast Touhoku district of Japan was struck by a set of massive earthquakes on the afternoon of March 11, 2011, which were soon followed by a huge tsunami that sent waves higher than 33 feet in the port city of Sendai 80 miles away and travelling at the speed of a jetliner. At nearly 9.0 on the Richter scale, it was one of the largest recorded earthquakes to hit Japan. It shifted the Earth's axis by 6 inches with an impact that was felt 250 miles inland in Tokyo, and which moved Eastern Japan 13 feet toward North America. Apart from huge loss of life and hazards of nuclear radiation arising from the crippled Daiichi Nuclear Reactors in Fukushima, the damage to the manufacturing plants in Japan exposed the hazards of interconnected global supply chains and their impact on factories located half way around the globe.

The impact of the earthquake was particularly acute on industries that rely on cutting edge electronic parts sourced from Japan. Shin-Etsu Chemical Company is the world's largest producer of silicon wafers and supplies 20 percent of the global capacity. Its centralized plant located 40 miles from the Fukushima nuclear facility was damaged in the earthquake, causing ripple effects at Intel and Toshiba that purchase wafers from Shin-Etsu. Similarly, a shortage of automotive sensors from Hitachi has slowed or halted production of vehicles in Germany, Spain, and France, while Chrysler is reducing overtime at factories in Mexico and Canada to conserve parts from Japan. Even worse, General Motors stopped production altogether at a plant in Louisiana and Ford closed a truck plant in Kentucky due to the quake. The supply of vehicles such as Toyota's Prius and Lexus will be limited in the United States because of production disruptions in its Japanese factories. China has been affected too, where ZTE Corporation is facing shortages of batteries and LCD screens for its cell phones. Similarly, Lenovo in China is looking at reduced supplies of components from Japan for assembly of its tablet computers. These disruptions due to reliance on small concentrated network of suppliers in Japan and globally connected production and logistics systems have caused worker layoffs an increase in prices of affected products, and economic losses that have been felt around the world.



Kimimasa Mayama/EPA/Landov

Following the strong earthquakes and tsunami, flames and smoke rise from a petroleum refining plant next to a heating power station in Shiogama, Miyagi Prefecture, northern Japan, about 220 km north of Tokyo.

Sources: Don Lee and David Pearson, "Disaster in Japan exposes supply chain weakness," *The State* (April 8, 2011), B6-B7; "Chrysler reduces overtime to help Japan," *The Associated Press* (April 8, 2011) printed in *The State* (April 6, 2011), B7; Krishna Dhir, "From the Editor," *Decision Line*, vol. 42, no. 2, 3.

skills required. In addition, when a firm's operations are scattered globally, customer response times can be longer. We discuss these issues in more depth in Chapter 10, "Designing Effective Supply Chains," because they should be considered when making decisions about outsourcing. Coordinating components from a wide array of suppliers can be challenging, as Nintendo found out in the production and worldwide distribution of its Wii game systems.[5] Despite twice increasing capacity since April 2007 to 1.8 million Wii's a month, Nintendo could only ship the completed units to retailers like Best Buy, Costco, and Circuit City in limited quantities that did not meet the large demand through the 2007 holiday season and beyond. In addition, as Managerial Practice 1.1 shows, catastrophic events such as the Japanese earthquake affect production and operations in Europe and United States because connected supply chains can spread disruptions rapidly and quickly across international borders.

Strong global competition affects industries everywhere. For example, U.S. manufacturers of steel, appliances, household durable goods, machinery, and chemicals have seen their market share decline in both domestic and international markets. With the value of world trade in services now at more than $2 trillion per year, banking, data processing, airlines, and consulting services are beginning to face many of the same international pressures. Regional trading blocs, such as EU and NAFTA, further change the competitive landscape in both services and manufacturing. Regardless of which area of the world you live in, the challenge is to produce services or products that can compete in a global market, and to design the processes that can make it happen.

Ethical, Workforce Diversity, and Environmental Issues

Businesses face more ethical quandaries than ever before, intensified by an increasing global presence and rapid technological change. As companies locate new operations and acquire more

[5] Peter Svensson, "GameStop to Sell Rain Checks for Wii," *The State* (December 18, 2007).



Weng lei/AP Photos

A Chinese consumer looks at Timberland products at a department store in Shanghai, China, November 11, 2010. Timberland seeks to benefit from rising incomes in the worlds fastest-growing major economy, and will also invest in its Hong Kong shops.

suppliers and customers in other countries, potential ethical dilemmas arise when business is conducted by different rules. Some countries are more sensitive than others about conflicts of interest, bribery, discrimination against minorities and women, minimum-wage levels, and unsafe workplaces. Managers must decide whether to design and operate processes that do more than just meet local standards. In addition, technological change brings debates about data protection and customer privacy. In an electronic world, businesses are geographically far from their customers, so a reputation of trust is paramount.

In the past, many people viewed environmental problems, such as toxic waste, poisoned drinking water, poor air quality, and climate change as quality-of-life issues; now, many people and businesses see them as survival issues. The automobile industry has seen innovation in electric and hybrid cars in response to environmental concerns and economic benefits arising from using less expensive fuels. Industrial nations face a particular burden because their combined populations consume proportionally much larger resources. Just seven nations, including the United States and Japan, produce almost half of all greenhouse gases. Now China and India have added to that total carbon footprint because of their vast economic and manufacturing expansion over the past decade.

Apart from government initiatives, large multinational companies have a responsibility as well for creating environmentally conscious practices, and can do so profitably. For instance, Timberland has over 110 stores in China because of strong demand for its boots, shoes, clothes, and outdoor gear in that country. It highlights its environmental credentials and corporate social responsibility through investments such as the reforestation efforts in northern China's Horqin Desert. Timberland hopes to double the number of stores over the next 3 years by environmentally differentiating itself from the competition. We discuss these issues in greater detail in Chapter 13, "Managing Sustainable Supply Chains."

The challenge is clear: Issues of ethics, workforce diversity, and the environment are becoming part of every manager's job. When designing and operating processes, managers should consider integrity, respect for the individual, and customer satisfaction along with more conventional performance measures such as productivity, quality, cost, and profit.

Operations Management as a Set of Decisions

In this text, we cover the major decisions operations managers make in practice. At the strategic level, operations managers are involved in the development of new capabilities and the maintenance of existing capabilities to best serve the firm's external customers. Operations managers design new processes that have strategic implications, and they are deeply involved in the development and organization of supply chains that link external suppliers and external customers to the firm's internal processes. Operations managers are often responsible for key performance measures such as cost and quality. These decisions have strategic impact because they affect the processes the firm uses to gain a competitive edge.

The operations manager's decisions should reflect corporate strategy. Plans, policies, and actions should be linked to those in other functional areas to support the firm's overall goals and objectives. These links are facilitated by taking a process view of a firm. Regardless of whether you aspire to be an operations manager, or you just want to use the principles of operations management to become a more effective manager, remember that effective management of people, capital, information, and materials is critical to the success of any process and any supply chain.

As you study operations management, keep two principles in mind:

1. Each part of an organization, not just the operations function, must design and operate processes that are part of a supply chain and deal with quality, technology, and staffing issues.
2. Each function of an organization has its own identity and yet is connected with operations through shared processes.

Great strategic decisions lead nowhere if the tactical decisions that support them are wrong. Operations managers are also involved in tactical decisions, including process improvement and performance measurement, managing and planning projects, generating production and staffing plans, managing inventories, and scheduling resources. You will find numerous examples of these decisions, and the implications of making them, throughout this text. You will also learn about

the decision-making tools practicing managers use to recognize and define the problem and then choose the best solution.

Computerized Decision-Making Tools

MyOMLab

MyOMLab contains a unique set of decision tools we call OM Explorer. This package contains powerful Excel-based computer routines to solve problems often encountered in practice. OM Explorer also has several tutors that provide coaching for all of the difficult analytical techniques in this text, and can be accessed from the drop-down menu. MyOMLab also contains POM for Windows, which is an extensive set of useful decision-making tools to complete your arsenal for solving operations problems, many Active Models (spreadsheets designed to help you learn more about important decision-making techniques), and a spreadsheet-based simulation package called SimQuick.

Addressing the Challenges in Operations Management

How can firms meet challenges today and in the future? One way is to recognize challenges as opportunities to improve existing processes and supply chains or to create new, innovative ones. The management of processes and supply chains goes beyond designing them; it requires the ability to ensure they achieve their goals. Firms should manage their processes and supply chains to maximize their competitiveness in the markets they serve. We share this philosophy of operations management, as illustrated in Figure 1.7. We use this figure at the start of each chapter to show how the topic of the chapter fits into our philosophy of operations management. In addition, this text also contains several chapter supplements that are not explicitly shown in Figure 1.7.

The figure shows that all effective operations decisions follow from a sound operations strategy. Consequently, our text has three major parts: "Part 1: Competing with Operations," Part 2: "Designing and Managing Processes," and "Part 3: Designing and Managing Supply Chains." The flow of topics reflects our approach of first understanding how a firm's operations can help provide a solid foundation for competitiveness before tackling the essential process design decisions that will support its strategies. Each part begins with a strategy discussion to support the decisions in that part. Once it is clear how firms design and improve processes, and how they implement those designs, we examine the design and operation of supply chains that link processes, whether they are internal or external to the firm. The performance of the supply chains determines the firm's outcomes, which include the services or products the firm produces, the financial results, and feedback from the firm's customers. These outcomes, which are considered in the firm's strategic plan, are discussed throughout this text.



▲ **FIGURE 1.7**
Managing Processes and Supply Chains

Part 1: Competing with Operations

The concluding chapter of Part 1 is a discussion of the methods and tools of project management. Project management is an effective approach to implementing operations strategy through the introduction of new services or products as well as any changes to a firm's processes or supply chains. Supplement A, "Decision Making Models," follows this chapter and covers some basic decision techniques that apply to multiple chapters.

Part 2: Designing and Managing Processes

In Part 2, we focus on analyzing processes and how they can be improved to meet the goals of the operations strategy. We begin by addressing the strategic aspects of process design and then present a six-step systematic approach to process analysis. Each chapter in this part deals with some aspect of that approach. We discuss the tools that help managers analyze processes, and we reveal the methods firms use to measure process performance and quality. These methods provide the foundation for programs such as Six Sigma and total quality management.

Determining the best process capacity with effective constraint management and making processes "lean" by eliminating activities that do not add value while improving those that do are also key decisions in the redesign of processes. The activities involved in managing processes are

essential for providing significant benefits to the firm. Effective management of its processes can allow a firm to reduce its costs and also increase customer satisfaction.

Part 3: Designing and Managing Supply Chains

The management of supply chains is based upon process management and operations strategy. In Part 2, we focus on individual processes. The focus of Part 3, however, is on supply chains involving processes both internal and external to the firm and the tools that enhance their execution. We begin this part with a look at managing inventory in supply chains. We follow that with understanding how the design of supply chains and major strategic decisions, such as outsourcing, inventory placement, and locating facilities affect performance. We also look at contemporary issues surrounding supply chain integration and the impact of supply chains on the environment. We follow that with chapters focused on three key planning activities for the effective operation of supply chains: (1) forecasting, (2) operations planning and scheduling, and (3) resource planning.

Adding Value with Process Innovation in Supply Chains

It is important to note that the effective operation of a firm and its supply chain is as important as the design and implementation of its processes. Process innovation can make a big difference even in a low-growth industry. Examining processes from the perspective of the value they add is an important part of a successful manager's agenda, as is gaining an understanding of how core processes and related supply chains are linked to their competitive priorities, markets, and the operations strategy of a firm. As illustrated by Zara in Managerial Practice 1.2, who says operations management does not make a difference?

MANAGERIAL PRACTICE 1.2 Zara–"Fast Fashion" Operational Innovation

Zara, the flagship brand of Inditex, was founded in 1963 by Amancio Ortega who started the business with 5,000 pesetas. In 1975, he opened his first Zara store in Spain. The innovation ethos that has always been consistent with Zara has been Ortega's view that modern women desire fashionable and good quality clothes, and Ortega has always believed that this requires operational innovations that will quickly translate catwalk and celebrity endorsed fashions into new in-demand clothes, which provide Zara with a competitive advantage over its rivals. By 2009, Zara had over 1,425 stores globally operating on either owned-joint venture or franchise arrangements. Its major presence remains within the European regions (78%) and the Americas (11% of stores), but it also has a presence in Middle East, Africa, and the Asia Pacific.

The principle that underpins "fast fashion" is the reduction of lead times to get the product from concept to consumer. Zara's products and clothes are recognized as being fashionable, meeting the current (in-demand) fashion trends and tastes for their customers. Frequently, these tastes may vary by geographical region with scope for country and even regional variances. Zara's global fashion retail operating geographical regions are: Europe, North and South America, The Middle East, Africa and the Asia Pacific.

As the clothes market is typically volatile due to quick changes in fashion, the Zara operational innovation allows for short life cycles, as the company understands that products will only be temporally fashionable. This is an important aspect of Zara's operations that typifies their continued success. The nature of Zara's market and likely order winners for Zara are that its designs are fashionable and are developed to meet fashion trends and tastes of customers who require fashion not durability. Zara's market is based on translating catwalk fashion quickly, while in-demand for a relatively short period of time, and for continuous replenishment with the next must-have fashion design.

Zara customers appreciate design exclusivity; they want a product with a short life cycle as this also contributes to its exclusivity. With this in mind, stores are stocked with new designs twice weekly, and collections are small and often sell out, which adds to the air of exclusivity and the reduction of the need for markdowns or sale items. Some of the operational innovations that have enable Zara to remain competitive for nearly 40 years is their vertical integrated business model which covers design, production, marketing and sales. Zara controls its entire production chain, whereas it main competitors Gap and H&M outsource all of their production.



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Zara's "fast-fashion" operational innovation and speed of response supply is typically 3 weeks compared to 3-9 months for the competition.

Zara's "fast fashion" operational innovation and speed of response supply is typically a remarkable 3 weeks in comparison to the competitors' 3–9 months. This provides the company with significant competitive advantages over their main rivals. It also ensures that production is more closely aligned to demand, in the right locations, while also keeping inventory markdowns and sale items to a minimum.

Source: Barnes, L. & Lea-Greenwood, G. (2010) Fast Fashion in the Retail Store Environment. International Journal of Retail & Distribution Management 38(10): 760–772.
Lopez, C. & Fan, Y. (2009) Internationalisation of the Spanish Fashion brand Zara. Journal of Fashion Marketing and Management 13(2): 279–296.
Rohwedder, C. & Johnson, K. (2008) Pace Setting Zara seeks more speed to fight its rising cheap-chic rivals. The Wall Street Journal. February (20) B1.

The topics in this text will help you meet operations challenges and achieve operational innovation regardless of your chosen career path.

LEARNING GOALS IN REVIEW

1. **Describe operations and supply chains in terms of inputs, processes, outputs, information flows, suppliers, and customers.** Review Figure 1.4 for the important supply chain linkage and information flows. The section "Operations and Supply Chain Management across the Organization," pp. 22–23, shows how different functional areas of business come together to create value for a firm.
2. **Define an operations strategy and its linkage to corporate strategy, as well as the role it plays as a source of competitive advantage in a global marketplace.** See the sections on "Operations Strategy" and "Corporate Strategy," pp. 27–30, and review Figure 1.5.
3. **Identify nine competitive priorities used in operations strategy, and their linkage to marketing strategy.** The section "Competitive Priorities and Capabilities," pp. 31–34, discusses the important concept of order winners and qualifiers. Review Table 1.1 for important illustrations and examples of how leading edge firms implemented different competitive priorities to create a unique positioning in the marketplace.
4. **Explain how operations can be used as a competitive weapon.** The section "Operations Strategy as a Pattern of Decisions," p. 35, shows how firms must identify gaps in their competitive priorities and build capabilities through related process and operational changes. Make sure that you review Table 1.3 that provides a nice illustrative example.
5. **Identify the global trends and challenges facing operations management.** The section "Trends in Operations Management," pp. 36–40, describes the pressures managers face for achieving productivity improvements, along with managing sustainability and work force diversity related issues in the face of global competition.

MyOMLab helps you develop analytical skills and assesses your progress with multiple problems on labor and multifactor productivity.

MyOMLab Resources	Titles	Link to the Book
Video	*Operations as a Competitive Weapon at Starwood*	A Process View; Operations Strategy as a Pattern of Decisions
Active Model Exercise	1.1 Productivity	Trends in Operations Management; Example 1.1 (pp. 36–37); Solved Problem 2 (pp. 44–45); Solved Problem 1 (p. 44)
OM Explorer Tutors	1.1 Productivity Measures	Trends in Operations Management; Example 1.1 (pp. 36–37); Solved Problem 2 (pp. 44–45); Solved Problem 1 (p. 44)
Tutor Exercises	1.1 Ticket sales	Example 1.1 (pp. 36–37); Solved Problem 2 (pp. 44–45); Solved Problem 1 (p. 44)
Virtual Tours	L'Oréal Cosmetics EDS Industries EDS Services	The Supply Chain View; Trends in Operations Management A Process View; The Supply Chain View A Process View; The Supply Chain View
Internet Exercises	1. Coca-Cola and Nestlé 2. Xerox 3. L'Oréal 4. Environment, Health, and Safety at Xerox	The Supply Chain View; Operations Strategy as a Pattern of Decisions The Supply Chain View; Competitive Priorities and Capabilities A Process View; Trends in Operations Management Trends in Operations Management
Additional Cases	BSB, Inc., The Pizza Wars Come to Campus	A Process View; The Supply Chain View; Competitive Priorities and Capabilities
Key Equations		
Image Library		

Key Equation

1. Productivity is the ratio of output to input:

$$\text{Productivity} = \frac{\text{Output}}{\text{Input}}$$

Key Terms

competitive capabilities 31
competitive priorities 31
consistent quality 32
core competencies 29
core process 27
customer relationship process 27
customization 32
delivery speed 32
development speed 32
external customers 24
external suppliers 24
internal customers 24
internal suppliers 24
lead time 29
low-cost operations 32
nested process 25
new service/product development process 27
on-time delivery 32
operation 22
operations management 22
operations strategy 27
order fulfillment process 27
order qualifier 33
order winner 31
process 22
productivity 36
supplier relationship process 27
supply chain 22
supply chain management 22
support process 27
time-based competition 31
top quality 32
variety 32
volume flexibility 32

Solved Problem 1

Natalie Attire makes fashionable garments. During a particular week, employees worked 360 hours to produce a batch of 132 garments, of which 52 were "seconds" (meaning that they were flawed). Seconds are sold for $90 each at Attire's Factory Outlet Store. The remaining 80 garments are sold to retail distribution at $200 each. What is the *labor* productivity ratio of this manufacturing process?

SOLUTION

$$\begin{aligned}\text{Value of output} &= (52 \text{ defective} \times 90/\text{defective}) + (80 \text{ garments} \times 200/\text{garment})\\ &= \$20{,}680\\ \text{Labor hours of input} &= 360 \text{ hours}\\ \text{Labor productivity} &= \frac{\text{Output}}{\text{Input}} = \frac{\$20{,}680}{360 \text{ hours}}\\ &= \$57.44 \text{ in sales per hour}\end{aligned}$$

Solved Problem 2

Student tuition at Boehring University is $150 per semester credit hour. The state supplements school revenue by $100 per semester credit hour. Average class size for a typical 3-credit course is 50 students. Labor costs are $4,000 per class, materials costs are $20 per student per class, and overhead costs are $25,000 per class.

a. What is the *multifactor* productivity ratio for this course process?

b. If instructors work an average of 14 hours per week for 16 weeks for each 3-credit class of 50 students, what is the *labor* productivity ratio?

SOLUTION

a. Multifactor productivity is the ratio of the value of output to the value of input resources.

$$\begin{aligned}\text{Value of output} &= \left(\frac{50 \text{ students}}{\text{class}}\right)\left(\frac{3 \text{ credit hours}}{\text{students}}\right)\left(\frac{\$150 \text{ tuition} + \$100 \text{ state support}}{\text{credit hour}}\right)\\ &= \$37{,}500/\text{class}\\ \text{Value of inputs} &= \text{Labor} + \text{Materials} + \text{Overhead}\\ &= \$4{,}000 + (\$20/\text{student} \times 50 \text{ students/class}) + \$25{,}000\\ &= \$30{,}000/\text{class}\\ \text{Multifactor productivity} &= \frac{\text{Output}}{\text{Input}} = \frac{\$37{,}500/\text{class}}{\$30{,}000/\text{class}} = 1.25\end{aligned}$$

b. Labor productivity is the ratio of the value of output to labor hours. The value of output is the same as in part (a), or \$37,500/class, so

$$\text{Labor hours of input} = \left(\frac{14 \text{ hours}}{\text{week}}\right)\left(\frac{16 \text{ weeks}}{\text{class}}\right) = 224 \text{ hours/class}$$

$$\text{Labor productivity} = \frac{\text{Output}}{\text{Input}} = \frac{\$37{,}500/\text{class}}{224 \text{ hours/class}} = \$167.41/\text{hour}$$

Discussion Questions

1. Wild West, Inc., is a regional telephone company that inherited nearly 100,000 employees and 50,000 retirees from AT&T. Wild West has a new mission: to diversify. It calls for a 10-year effort to enter the financial services, real estate, cable TV, home shopping, entertainment, and cellular communication services markets—and to compete with other telephone companies. Wild West plans to provide cellular and fiber-optic communications services in markets with established competitors, such as the United Kingdom, and in markets with essentially no competition, such as Russia and former Eastern Bloc countries.

- **a.** What environmental forces could be at work that Wild West should consider?
- **b.** What types of strategic plans must Wild West make? Is the "do-nothing" option viable? If Wild West's mission appears too broad, which businesses would you trim first?
- **c.** What are the possible core competencies of Wild West? What weaknesses should it avoid or mitigate?

2. Consider amazon.com, whose Web site enjoys millions of "hits" each day and puts customers in touch with millions of services and products. What are amazon.com's competitive priorities and what should its operations strategy focus on?

3. Although all nine of the competitive priorities discussed in this chapter are relevant to a company's success in the marketplace, explain why a company should not necessarily try to excel in all of them. What determines the choice of the competitive priorities that a company should emphasize for its key processes?

4. A local fast-food restaurant processes several customer orders at once. Service clerks cross paths, sometimes nearly colliding, while they trace different paths to fill customer orders. If customers order a special combination of toppings on their hamburgers, they must wait quite some time while the special order is cooked. How would you modify the restaurant's operations to achieve competitive advantage? Because demand surges at lunchtime, volume flexibility is a competitive priority in the fast-food business. How would you achieve volume flexibility?

5. Suppose that you were conducting a market analysis for a new textbook about technology management. What would you need to know to identify a market segment? How would you make a needs assessment? What should be the collection of services and products?

6. A local hospital declares that it is committed to provide *care* to patients arriving at the emergency unit in less than 15 minutes and that it will never turn away patients who need to be hospitalized for further medical care. What implications does this commitment have for strategic operations management decisions (i.e., decisions relating to capacity and workforce)?

7. You are designing a grocery delivery business. Via the Internet, your company will offer staples and frozen foods in a large metropolitan area and then deliver them within a customer-defined window of time. You plan to partner with two major food stores in the area. What should be your competitive priorities and what capabilities do you want to develop in your core and support processes?

8. FedEx built its business on quick, dependable delivery of items being shipped by air from one business to another. Its early advantages included global tracking of shipments using Web technology. The advancement of Internet technology enabled competitors to become much more sophisticated in order tracking. In addition, the advent of Web-based businesses put pressure on increased ground transportation deliveries. Explain how this change in the environment has affected FedEx's operations strategy, especially relative to UPS, which has a strong hold on the business-to-consumer ground delivery business.

9. Kathryn Shoemaker established Grandmother's Chicken Restaurant in Middlesburg 5 years ago. It features a unique recipe for chicken, "just like grandmother used to make." The facility is homey, with relaxed and friendly service. Business has been good during the past 2 years, for both lunch and dinner. Customers normally wait about 15 minutes to be served, although complaints about service delays have increased recently. Shoemaker is currently considering whether to expand the current facility or open a similar restaurant in neighboring Uniontown, which has been growing rapidly.

- **a.** What are the possible distinctive competencies of Grandmother's?
- **b.** What environmental forces could be at work in Middlesburg and Uniontown that Shoemaker should consider?
- **c.** What types of strategic plans must Shoemaker make?

10. Consider your last (or current) job.

- **a.** What activities did you perform?
- **b.** Was your position in accounting, finance, human resources, management information systems, marketing, operations, or other? Explain.
- **c.** How could you measure the customer value you were adding by performing your activities?
- **d.** Who were your customers (internal and external), and how did you interact with them?

11. Choosing which processes are core to a firm's competitive position is a key strategic decision. For example, Nike, a popular sports shoe company, focuses on the customer relationship, new product development, and supplier relationship processes and leaves the order fulfillment process to others. Allen Edmonds, a top-quality shoe company, considers all four processes to be core processes. What considerations would you make in determining which processes should be core to your manufacturing company?

Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decision, the software entirely replaces the manual calculations.

1. Alyssa's Custom Cakes currently sells 5 birthday, 2 wedding, and 3 specialty cakes each month for $50, $150, and $100 each, respectively. The cost of labor is $50 per hour including benefits. It takes 90 minutes to produce a birthday cake, 240 minutes to produce a wedding cake, and 60 minutes to produce a specialty cake. Alyssa's current multifactor productivity ratio is 1.25.

a. Use the multifactor productivity ratio provided to calculate the average cost of the cakes produced.

b. Based on your answer in part (a), is there a type of cake Alyssa should stop selling?

c. Based solely on the labor productivity ratio, which cake should Alyssa try to sell the most?

d. Calculate Alyssa's labor productivity ratio in dollars per hour for each type of cake.

2. Suds and Duds Laundry washed and pressed the following numbers of dress shirts per week.

Week	Work Crew	Total Hours	Shirts
1	Sud and Dud	24	68
2	Sud and Jud	46	130
3	Sud, Dud, and Jud	62	152
4	Sud, Dud, and Jud	51	125
5	Dud and Jud	45	131

a. Calculate the *labor* productivity ratio for each week.

b. Explain the labor productivity pattern exhibited by the data.

3. The output of a process is valued at $100 per unit. The cost of labor is $50 per hour including benefits. The accounting department provided the following information about the process for the past four weeks:

	Week 1	Week 2	Week 3	Week 4
Units Produced	1,124	1,310	1,092	981
Labor ($)	12,735	14,842	10,603	9,526
Material ($)	21,041	24,523	20,442	18,364
Overhead ($)	8,992	10,480	8,736	7,848

a. Use the multifactor productivity ratio to see whether recent process improvements had any effect and, if so, when the effect was noticeable.

b. Has labor productivity changed? Use the labor productivity ratio to support your answer.

4. CD players are produced on an automated assembly line process. The standard cost of CD players is $150 per unit (labor, $30; materials, $70; and overhead, $50). The sales price is $300 per unit.

a. To achieve a 10 percent multifactor productivity improvement by reducing materials costs only, by what percentage must these costs be reduced?

b. To achieve a 10 percent multifactor productivity improvement by reducing labor costs only, by what percentage must these costs be reduced?

c. To achieve a 10 percent multifactor productivity improvement by reducing overhead costs only, by what percentage must these costs be reduced?

5. (Refer to Solved Problem 2.) Coach Bjourn Toulouse led the Big Red Herrings to several disappointing football seasons. Only better recruiting will return the Big Red Herrings to winning form. Because of the current state of the program, Boehring University fans are unlikely to support increases in the $192 season ticket price. Improved recruitment will increase overhead costs to $30,000 per class section from the current $25,000 per class section. The university's budget plan is to cover recruitment costs by increasing the average class size to 75 students. Labor costs will increase to $6,500 per 3-credit course. Material costs will be about $25 per student for each 3-credit course. Tuition will be $200 per semester credit, which is supplemented by state support of $100 per semester credit.

a. What is the multifactor productivity ratio? Compared to the result obtained in Solved Problem 2, did productivity increase or decrease for the course process?

b. If instructors work an average of 20 hours per week for 16 weeks for each 3-credit class of 75 students, what is the *labor* productivity ratio?

Advanced Problems

6. Mariah Enterprises makes a variety of consumer electronic products. Its camera manufacturing plant is considering choosing between two different processes, named Alpha and Beta, which can be used to make a component part. To make the correct decision, the managers would like to compare the labor and multifactor productivity of process Alpha with that of process Beta. The value of process output for Alpha and Beta is $175 and $140 per unit, and the corresponding overhead costs are $6,000 and $5,000, respectively.

	PROCESS ALPHA		PROCESS BETA	
Product	**A**	**B**	**A**	**B**
Output (units)	50	60	30	80
Labor ($)	$1,200	$1,400	$1,000	$2,000
Material ($)	$2,500	$3,000	$1,400	$3,500

a. Which process, Alpha or Beta, is more productive?

b. What conclusions can you draw from your analysis?

7. Mack's guitar fabrication shop produces low-cost, highly durable guitars for beginners. Typically, out of the 100 guitars that begin production each month, only 80 percent are considered good enough to sell. The other 20 percent are scrapped due to quality problems that are identified after they have completed the production process. Each guitar sells for $250. Because some of the production process is automated, each guitar only requires 10 labor hours. Each employee works an average 160 hours per month. Labor is paid at $10/hour, materials cost is $40/guitar, and overhead is $4,000.

a. Calculate the labor and multifactor productivity ratios.

b. After some study, the operations manager Darren Funk recommends three options to improve the company's multifactor productivity: (1) increase the sales price by 10 percent, (2) improve quality so that only 10 percent are defective, or (3) reduce labor, material, and overhead costs by 10 percent. Which option has the greatest impact on the multifactor productivity measure?

8. The Big Black Bird Company (BBBC) has a large order for special plastic-lined military uniforms to be used in an urgent military operation. Working the normal two shifts of 40 hours each per week, the BBBC production process usually produces 2,500 uniforms per week at a standard cost of $120 each. Seventy employees work the first shift and 30 employees work the second. The contract price is $200 per uniform. Because of the urgent need, BBBC is authorized to use around-the-clock production, 6 days per week. When each of the two shifts works 72 hours per week, production increases to 4,000 uniforms per week but at a cost of $144 each.

a. Did the multifactor productivity ratio increase, decrease, or remain the same? If it changed, by what percentage did it change?

b. Did weekly profits increase, decrease, or remain the same?

c. Did the labor productivity ratio increase, decrease, or remain the same? If it changed, by what percentage did it change?

9. The Morning Brew Coffee Shop sells Regular, Cappuccino, and Vienna blends of coffee. The shop's current daily labor cost is $320, the equipment cost is $125, and the overhead cost is $225. Daily demands, along with selling price and material costs per beverage, are given below.

	Regular Coffee	**Cappuccino**	**Vienna coffee**
Beverages sold	350	100	150
Price per beverage	$2.00	$3.00	$4.00
Material ($)	$0.50	$0.75	$1.25

Harald Luckerbauer, the manager at Morning Brew Coffee Shop, would like to understand how adding Eiskaffee (a German coffee beverage of chilled coffee, milk, sweetener, and vanilla ice cream) will alter the shop's productivity. His market research shows that Eiskaffee will bring in new customers and not cannibalize current demand. Assuming that the new equipment is purchased before Eiskaffee is added to the menu, Harald has developed new average daily demand and cost projections. The new equipment cost is $200, and the overhead cost is $350. Modified daily demands, as well as selling price and material costs per beverage for the new product line, are given below.

	Regular Coffee	**Cappuccino**	**Vienna coffee**	**Eiskaffee**
Beverages sold	350	100	150	75
Price per beverage	$2.00	$3.00	$4.00	$5.00
Material ($)	$0.50	$0.75	$1.25	$1.50

a. Calculate the change in labor and multifactor productivity if Eiskaffee is added to the menu.

b. If everything else remains unchanged, how many units of Eiskaffee would have to be sold to ensure that the multifactor productivity increases from its current level?

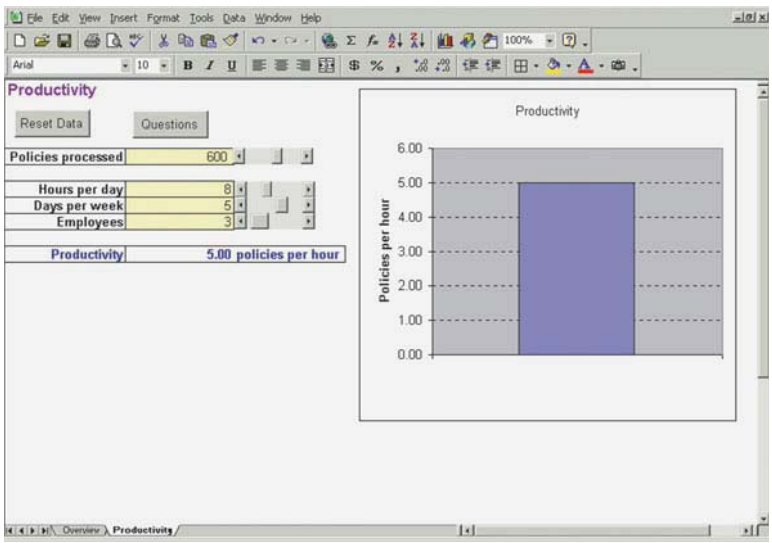
Active Model Exercise

This Active Model appears in MyOMLab. It allows you to evaluate the important elements of labor productivity.

QUESTIONS

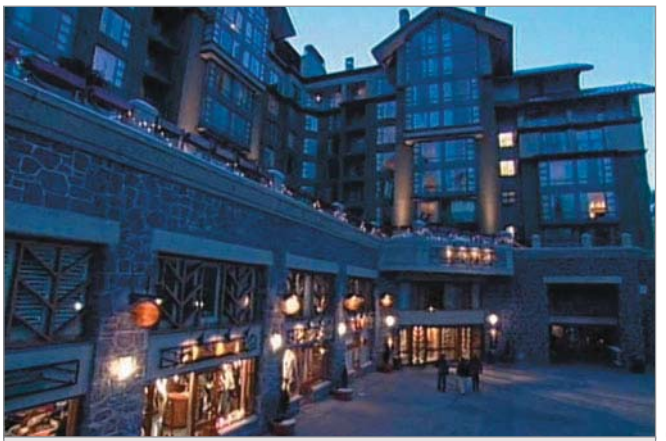
1. If an employee is hired, what will be the weekly number of policies processed if the productivity of five policies per hour is maintained?
2. Suppose that, during the summer, the company works for only 4 days per week. What will be the weekly number of policies processed if the productivity of five policies per hour is maintained?
3. If the insurance company can process 60 (10 percent) more policies per week, by what percentage will the productivity measure rise?
4. Suppose the 8-hour day includes a 45-minute lunch. What is the revised productivity measure, excluding lunch?

▶ **ACTIVE MODEL 1.1**
Labor Productivity Using Data from Example 1.1



VIDEO CASE Operations as a Competitive Weapon at Starwood

Starwood is one of the world's largest hotel companies, with more than 750 owned, managed, and franchised properties in more than 80 countries. The company's lodging brands include The Luxury Collection, St. Regis, Sheraton, Westin, Four Points, and W Hotels. Its hotels regularly appear on lists of top hotels around the world. On any given night, guests in the hotels may be individual leisure travelers, independent business guests, or part of a meeting or convention.



Pearson

In 2002, Starwood standardized its operating processes so that it could measure, improve, and ultimately grow its convention business. Each meeting is assigned a Star Meeting Concierge who works closely with meeting planners.

When guests stay at a Starwood property as part of a meeting or convention, arrangements are typically made by a meeting planner. The meeting planner works with a location to arrange meeting facilities, banquet rooms, lodging, and events for participants. Prior to 2002, the company's individual properties had their own approaches to convention planning, yet no consistent, coordinated program within or across brands made it easy for meeting planners to do business with Starwood. For example, paperwork for confirming program details, rooms, and food and beverage requirements differed between properties and brands. Some hotels had diagrams of meeting space, while others did not. Technology available for meeting rooms varied widely, and a hotel liaison was not always immediately available during the event in case a need arose.

Recognizing that Starwood's future growth and success relied heavily on its relationships with meeting planners, the company held focus groups to gather information about their needs and expectations. One clear priority emerged: consistency in the meeting planning process, whether that meeting was held at the Sheraton in New York, the Westin Kierland in Phoenix, or the W Hotel in Lakeshore, Chicago. Such a program could create consistency across all brands, and generate loyalty and increased revenues from those meeting planners who drive large volumes of business to Starwood properties annually.

As a result of the meetings, Starwood created the Starwood Preferred Planner program. Every hotel property now has the same paperwork for the meeting planning process, and shares that paperwork electronically across properties and brands. Contracts were standardized and new standards created to recognize and reward frequent VIP meeting planners. Each meeting is assigned a "Star Meeting Concierge" whose sole responsibility is to anticipate and fulfill any needs of the meeting planner during the event. Handheld Nextel radio phones are now issued at check-in to the meeting planners at no extra charge so that they have 24-hour access to the concierge.

To measure the performance of the new process, Starwood set high internal targets for scores on the surveys given to meeting planners after their events concluded. For instance, at the Luxury Collection and St. Regis brands, individual meeting scores must be 4.55 on a 5-point scale. At the Westin and W Hotels, scores must be above 4.35 on the 5-point scale. Scores from Sheraton properties must exceed 4.30, and Four Points hotels have a target of 4.25 on the 5-point scale. Because the expectations for an airport location one-day meeting (not held at the St. Regis or Luxury Collection) differ from a multiday resort experience, the targets reflect those expectations.

QUESTIONS

1. What are the key inputs and outputs associated with Starwood's new meeting planning process?
2. How does the meeting planning process at Starwood interact with the following core processes in their hotels?
   - **a.** Customer relationship (internal and external)
   - **b.** New service or product development
   - **c.** Order fulfillment
   - **d.** Supplier relationship

CASE Theorganicgrocer.com

Theorganicgrocer.com is an online supplier of organically produced fruit and vegetables. Clients within a 50 km radius of Bangor in Wales are promised next-day delivery at a time and place selected by the customer. Maria Heales, a joint owner says, "Our business has been based on market research.We've done a great deal of background work to ensure our business model works. All the early signs suggested that people would order online if the quality was good, payment was secure, the price was right and delivery could be trusted…. 'Dependability, reliability, timeliness, quality and price' were the words that sung out," she says. The service offering includes delivery, 9 A.M. to 9 P.M., six days a week; organic produce; and customer-specified date and time of delivery. All produce is direct from the market on the day of delivery from a chilled vehicle. The Web site is updated daily to reflect current price and in-season produce. "The key to our success," Maria says, "is the simplicity of use of our Web site."

Vehicle route planning ensures deliveries are optimized, and aggregated stock ordering means only produce previously ordered by customers is purchased. Customers' payments are made online using a secure, encrypted facility administered by a national bank. "But our business has grown, so we need to expand capacity and capability.We are no longer a fruit and 'veg' stall online!" Maria's husband and business partner, Marcus, says. Their goal now is to expand their business to include regional franchising, product range expansion and, for example, birthday gifts, filled baskets, and ready-to-eat

gourmet selections. Operating from a small "lock-up" unit on the fringe of town, expansion of the business would require considerable capital investment. Moreover, the operations of the future business will be more complex.

Currently, 95 percent of the customers are households, with the remainder being local restaurants. Approximately 20 percent of their customer base generates 80 percent of the profit. Theorganicgrocer.com has a financial system—a computer application that Marcus modified to meet its needs. A separate computer package generates advertising material and mail-outs. This tends to be done by Maria when time permits. However, there is little market information to support strategic operations planning. Much of Maria and Marcus's aspiration is based on the success of their current operations and the hope that the trend continues. Whilst there are several competitors supplying online vegetables, they do not offer organic products and they have limited products. They are, however, some 20 percent cheaper than Theorganicgrocer.com.

As business has grown, there have been complaints of damaged products, and growth in customers has caused the average on-time deliveries (plus or minus 20 minutes) to fall from 95 percent to 75 percent. All this, Maria and Marcus argue, is more reason for expanding the business.

Marcus enjoys the technical aspects of the online business, and spends much of his time modifying and improving the Web site. When necessary, he helps to consolidate customers' orders. "I think my time is better spent on Web site development," Marcus says; "It's critical for the business if it goes down or it isn't showing current product information, then sales are jeopardized." Presently, Maria takes responsibility for placing orders to their thirty suppliers. Maria and the delivery drivers pack customer orders into salvaged cardboard boxes that are of various sizes and often damaged. "But at least they are free!" she laughs.

QUESTIONS

1. What are the comparative operational differences of a conventional "bricks and mortar" grocery to an online operation? What are the comparative value chains for each system?
2. What are the key processes in the online operation?
3. Where are the possible operational problem areas of their online business, now and in the future?
4. What operational improvements need to be taken now to improve competitiveness?

Source: This fictional case was prepared by Dr. David Parker, The University of Queensland, Business School, as a basis of classroom discussion.

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SUPPLEMENT

A

DECISION MAKING MODELS

Operations managers make many decisions as they manage processes and supply chains. Although the specifics of each situation vary, decision making generally involves the same basic steps: (1) recognize and clearly define the problem, (2) collect the information needed to analyze possible alternatives, and (3) choose and implement the most feasible alternative.

Sometimes, hard thinking in a quiet room is sufficient. At other times, interacting with others or using more formal procedures are needed. Here, we present four such formal procedures: break-even analysis, the preference matrix, decision theory, and the decision tree.

- Break-even analysis helps the manager identify how much change in volume or demand is necessary before a second alternative becomes better than the first alternative.
- The preference matrix helps a manager deal with multiple criteria that cannot be evaluated with a single measure of merit, such as total profit or cost.
- Decision theory helps the manager choose the best alternative when outcomes are uncertain.
- A decision tree helps the manager when decisions are made sequentially—when today's best decision depends on tomorrow's decisions and events.

Break-Even Analysis

To evaluate an idea for a new service or product, or to assess the performance of an existing one, determining the volume of sales at which the service or product breaks even is useful. The **break-even quantity** is the volume at which total revenues equal total costs. Use of this technique is known as **break-even analysis**. Break-even analysis can also be used to compare processes by finding the volume at which two different processes have equal total costs.

break-even quantity

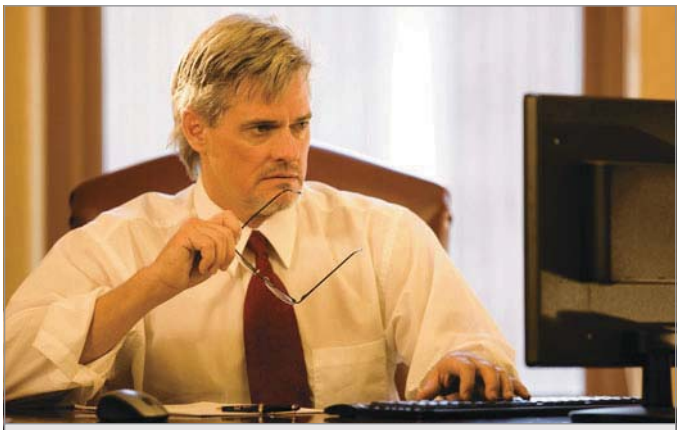
The volume at which total revenues equal total costs.

break-even analysis

The use of the break-even quantity; it can be used to compare processes by finding the volume at which two different processes have equal total costs.

LEARNING GOALS *After reading this supplement, you should be able to:*

1. Explain break-even analysis, using both the graphic and algebraic approaches.
2. Define a preference matrix.
3. Explain how to construct a payoff table.
4. Identify the maximin, maximax, Laplace, minimax regret, and expected value decision rules.
5. Describe how to draw and analyze a decision tree.



Sportstock/Shutterstock

A manager is doing some hard thinking and analysis on his computer before reaching a final decision.

Evaluating Services or Products

We begin with the first purpose: to evaluate the profit potential of a new or existing service or product. This technique helps the manager answer questions, such as the following:

- Is the predicted sales volume of the service or product sufficient to break even (neither earning a profit nor sustaining a loss)?
- How low must the variable cost per unit be to break even, based on current prices and sales forecasts?
- How low must the fixed cost be to break even?
- How do price levels affect the break-even volume?

Break-even analysis is based on the assumption that all costs related to the production of a specific service or product can be divided into two categories: (1) variable costs and (2) fixed costs.

variable cost

The portion of the total cost that varies directly with volume of output.

fixed cost

The portion of the total cost that remains constant regardless of changes in levels of output.

The **variable cost**, c, is the portion of the total cost that varies directly with volume of output: costs per unit for materials, labor, and usually some fraction of overhead. If we let Q equal the number of customers served or units produced per year, total variable cost = cQ. The **fixed cost**, F, is the portion of the total cost that remains constant regardless of changes in levels of output: the annual cost of renting or buying new equipment and facilities (including depreciation, interest, taxes, and insurance); salaries; utilities; and portions of the sales or advertising budget. Thus, the total cost of producing a service or good equals fixed costs plus variable costs multiplied by volume, or

$$\text{Total cost} = F + cQ$$

The variable cost per unit is assumed to be the same no matter how small or large Q is, and thus, total cost is linear. If we assume that all units produced are sold, total annual revenues equal revenue per unit sold, p, multiplied by the quantity sold, or

$$\text{Total revenue} = pQ$$

If we set total revenue equal to total cost, we get the break-even quantity point as

$$\begin{aligned} pQ &= F + cQ \\ (p - c)Q &= F \\ Q &= \frac{F}{p - c} \end{aligned}$$

We can also find this break-even quantity graphically. Because both costs and revenues are linear relationships, the break-even quantity is where the total revenue line crosses the total cost line.

EXAMPLE A.1 Finding the Break-Even Quantity

MyOMLab
Active Model A.1 in MyOMLab provides additional insight on this break-even example and its extensions with four "what-if" questions.

A hospital is considering a new procedure to be offered at $200 per patient. The fixed cost per year would be $100,000, with total variable costs of $100 per patient. What is the break-even quantity for this service? Use both algebraic and graphic approaches to get the answer.

SOLUTION

The formula for the break-even quantity yields

$$Q = \frac{F}{p - c} = \frac{100,000}{200 - 100} = 1,000 \text{ patients}$$

MyOMLab
Tutor A.1 in MyOMLab provides a new example to practice break-even analysis.

To solve graphically we plot two lines: one for costs and one for revenues. Two points determine a line, so we begin by calculating costs and revenues for two different output levels. The following table shows the results for Q = 0 and Q = 2,000. We selected zero as the first point because of the ease of plotting total revenue (0) and total cost (F). However, we could have used any two reasonably spaced output levels.

Quantity (patients) (Q)	Total Annual Cost ($) (100,000 + 100$Q$)	Total Annual Revenue ($) (200$Q$)
0	100,000	0
2,000	300,000	400,000

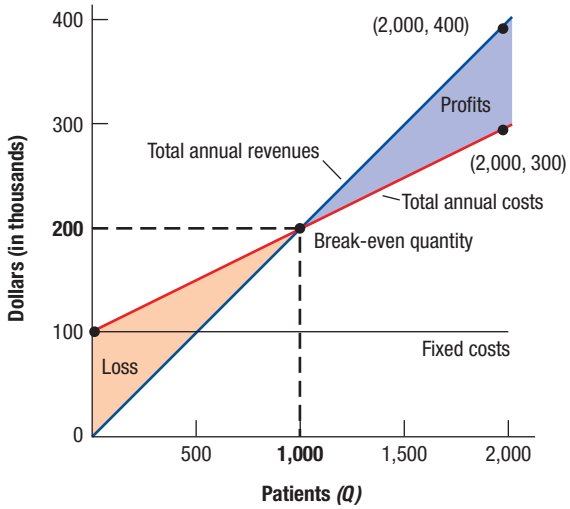


FIGURE A.1
Graphic Approach to Break-Even Analysis

We can now draw the cost line through points (0, 100,000) and (2,000, 300,000). The revenue line goes between (0, 0) and (2,000, 400,000). As Figure A.1 indicates, these two lines intersect at 1,000 patients, the break-even quantity.

DECISION POINT

Management expects the number of patients needing the new procedure will exceed the 1,000-patient break-even quantity but first wants to learn how sensitive the decision is to demand levels before making a final choice.

Break-even analysis cannot tell a manager whether to pursue a new service or product idea or drop an existing line. The technique can only show what is likely to happen for various forecasts of costs and sales volumes. To evaluate a variety of "what-if" questions, we use an approach called **sensitivity analysis**, a technique for systematically changing parameters in a model to determine the effects of such changes. The concept can be applied later to other techniques, such as linear programming. Here we assess the sensitivity of total profit to different pricing strategies, sales volume forecasts, or cost estimates.

sensitivity analysis
A technique for systematically changing parameters in a model to determine the effects of such changes.

EXAMPLE A.2 Sensitivity Analysis of Sales Forecasts

If the most pessimistic sales forecast for the proposed service in Figure A.1 were 1,500 patients, what would be the procedure's total contribution to profit and overhead per year?

SOLUTION

The graph shows that even the pessimistic forecast lies above the break-even volume, which is encouraging. The procedure's total contribution, found by subtracting total costs from total revenues, is

$$pQ - (F + cQ) = 200(1{,}500) - [100{,}000 + 100(1{,}500)]$$
$$= \$50{,}000$$

DECISION POINT

Even with the pessimistic forecast, the new procedure contributes $50,000 per year. After evaluating the proposal with present value method (see MyOMLab Supplement F), management added the new procedure to the hospital's services.

MyOMLab

Evaluating Processes

Often, choices must be made between two processes or between an internal process and buying services or materials on the outside. In such cases, we assume that the decision does not affect revenues. The manager must study all the costs and advantages of each approach. Rather than find the quantity at which total costs equal total revenues, the analyst finds the quantity for which the total costs for

two alternatives are equal. For the make-or-buy decision, it is the quantity for which the total "buy" cost equals the total "make" cost. Let F_b equal the fixed cost (per year) of the buy option, F_m equal the fixed cost of the make option, c_b equal the variable cost (per unit) of the buy option, and c_m equal the variable cost of the make option. Thus, the total cost to buy is $F_b + c_bQ$ and the total cost to make is $F_m + c_mQ$. To find the break-even quantity, we set the two cost functions equal and solve for Q:

$$F_b + c_bQ = F_m + c_mQ$$

$$Q = \frac{F_m - F_b}{c_b - c_m}$$

The make option should be considered, ignoring qualitative factors, only if its variable costs are lower than those of the buy option. The reason is that the fixed costs for making the service or product are typically higher than the fixed costs for buying. Under these circumstances, the buy option is better if production volumes are less than the break-even quantity. Beyond that quantity, the make option becomes better. Chapter 10, "Designing Effective Supply Chains," brings out other considerations when making make-or-buy decisions.

EXAMPLE A.3 Break-Even Analysis for Make-or-Buy Decisions

MyOMLab
Active Model A.2 in MyOMLab provides additional insight on this make-or-buy example and its extensions.

The manager of a fast-food restaurant featuring hamburgers is adding salads to the menu. For each of the two new options, the price to the customer will be the same. The make option is to install a salad bar stocked with vegetables, fruits, and toppings and let the customer assemble the salad. The salad bar would have to be leased and a part-time employee hired. The manager estimates the fixed costs at $12,000 and variable costs totaling $1.50 per salad. The buy option is to have preassembled salads available for sale. They would be purchased from a local supplier at $2.00 per salad. Offering preassembled salads would require installation and operation of additional refrigeration, with an annual fixed cost of $2,400. The manager expects to sell 25,000 salads per year.

What is the make-or-buy quantity?

MyOMLab
Tutor A.2 in MyOMLab provides a new example to practice break-even analysis on make-or-buy decisions.

SOLUTION

The formula for the break-even quantity yields the following:

$$Q = \frac{F_m - F_b}{c_b - c_m} = \frac{12,000 - 2,400}{2.0 - 1.5} = 19,200 \text{ salads}$$

FIGURE A.2 ▶
Break-Even Analysis Solver of OM Explorer for Example A.3

	Process 1	Process 2
Fixed costs (F)	$12,000	$2,400
Variable costs (c)	$1.50	$2.00
Expected demand	25,000	
Break-even quantity	19,200.0	
Decision: Process 1		

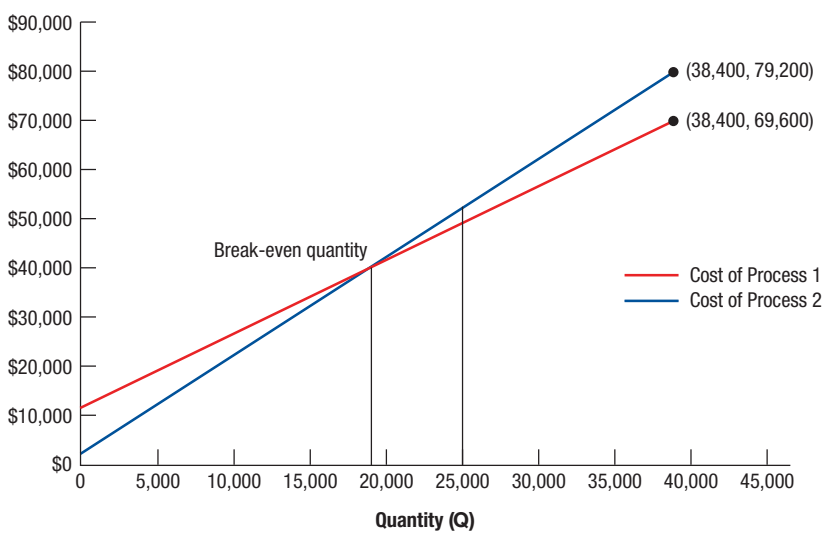


Figure A.2 shows the solution from OM Explorer's *Break-Even Analysis* Solver. The break-even quantity is 19,200 salads. As the 25,000-salad sales forecast exceeds this amount, the make option is preferred. Only if the restaurant expected to sell fewer than 19,200 salads would the buy option be better.

DECISION POINT

Management chose the make option after considering other qualitative factors, such as customer preferences and demand uncertainty. A deciding factor was that the 25,000-salad sales forecast is well above the 19,200-salad break-even quantity.

Preference Matrix

Decisions often must be made in situations where multiple criteria cannot be naturally merged into a single measure (such as dollars). For example, a manager deciding in which of two cities to locate a new plant would have to consider such unquantifiable factors as quality of life, worker attitudes toward work, and community reception in the two cities. These important factors cannot be ignored. A **preference matrix** is a table that allows the manager to rate an alternative according to several performance criteria. The criteria can be scored on any scale, such as from 1 (worst possible) to 10 (best possible) or from 0 to 1, as long as the same scale is applied to all the alternatives being compared. Each score is weighted according to its perceived importance, with the total of these weights typically equaling 100. The total score is the sum of the weighted scores (weight × score) for all the criteria. The manager can compare the scores for alternatives against one another or against a predetermined threshold.

preference matrix

A table that allows the manager to rate an alternative according to several performance criteria.

EXAMPLE A.4 Evaluating an Alternative with a Preference Matrix

The following table shows the performance criteria, weights, and scores (1 = worst, 10 = best) for a new product: a thermal storage air conditioner. If management wants to introduce just one new product and the highest total score of any of the other product ideas is 800, should the firm pursue making the air conditioner?

MyOMLab
Tutor A.3 in MyOMLab provides a new example to practice with preference matrixes.

Performance Criterion	Weight *(A)*	Score *(B)*	Weighted Score *(A × B)*
Market potential	30	8	240
Unit profit margin	20	10	200
Operations compatibility	20	6	120
Competitive advantage	15	10	150
Investment requirement	10	2	20
Project risk	5	4	20
			Weighted score = 750

SOLUTION

Because the sum of the weighted scores is 750, it falls short of the score of 800 for another product. This result is confirmed by the output from OM Explorer's *Preference Matrix* Solver in Figure A.3.

Insert a Criterion | Add a Criterion | Remove a Criterion

	Weight (A)	Score (B)	Weighted Score (A x B)
Market potential	30	8	240
Unit profit margin	20	10	200
Operations compatability	20	6	120
Competitive advantage	15	10	150
Investment requirement	10	2	20
Project risk	5	4	20
	Final Weighted Score		750

◀ **FIGURE A.3**
Preference Matrix Solver for Example A.4

DECISION POINT

Management should drop the thermal storage air-conditioner idea. Another new product idea is better, considering the multiple criteria, and management only wanted to introduce one new product at the time.

Not all managers are comfortable with the preference matrix technique. It requires the manager to state criteria weights before examining the alternatives, although the proper weights may not be readily apparent. Perhaps, only after seeing the scores for several alternatives can the manager decide what is important and what is not. Because a low score on one criterion can be compensated for or overridden by high scores on others, the preference matrix method also may cause managers to ignore important signals. In Example A.4, the investment required for the thermal storage air conditioner might exceed the firm's financial capability. In that case, the manager should not even be considering the alternative, no matter how high its score.

Decision Theory

decision theory

A general approach to decision making when the outcomes associated with alternatives are often in doubt.

Decision theory is a general approach to decision making when the outcomes associated with alternatives are often in doubt. It helps operations managers with decisions on process, capacity, location, and inventory because such decisions are about an uncertain future. Decision theory can also be used by managers in other functional areas. With decision theory, a manager makes choices using the following process:

1. List the feasible *alternatives*. One alternative that should always be considered as a basis for reference is to do nothing. A basic assumption is that the number of alternatives is finite. For example, in deciding where to locate a new retail store in a certain part of the city, a manager could theoretically consider every grid coordinate on the city's map. Realistically, however, the manager must narrow the number of choices to a reasonable number.
2. List the *events* (sometimes called *chance events* or *states of nature*) that have an impact on the outcome of the choice but are not under the manager's control. For example, the demand experienced by the new facility could be low or high, depending not only on whether the location is convenient to many customers, but also on what the competition does and general retail trends. Then, group events into reasonable categories. For example, suppose that the average number of sales per day could be anywhere from 1 to 500. Rather than have 500 events, the manager could represent demand with just 3 events: 100 sales/day, 300 sales/day, or 500 sales/day. The events must be mutually exclusive and collectively exhaustive, meaning that they do not overlap and that they cover all eventualities.

payoff table

A table that shows the amount for each alternative if each possible event occurs.

3. Calculate the *payoff* for each alternative in each event. Typically, the payoff is total profit or total cost. These payoffs can be entered into a **payoff table**, which shows the amount for each alternative if each possible event occurs. For 3 alternatives and 4 events, the table would have 12 payoffs (3×4). If significant distortions will occur if the time value of money is not recognized, the payoffs should be expressed as present values or internal rates of return (see MyOMLab Supplement F.) For multiple criteria with important qualitative factors, use the weighted scores of a preference matrix approach as the payoffs.

MyOMLab

4. Estimate the likelihood of each event, using past data, executive opinion, or other forecasting methods. Express it as a *probability*, making sure that the probabilities sum to 1.0. Develop probability estimates from past data if the past is considered a good indicator of the future.
5. Select a *decision rule* to evaluate the alternatives, such as choosing the alternative with the lowest expected cost. The rule chosen depends on the amount of information the manager has on the event probabilities and the manager's attitudes toward risk.

Using this process, we examine decisions under three different situations: certainty, uncertainty, and risk.

Decision Making under Certainty

The simplest situation is when the manager knows which event will occur. Here the decision rule is to pick the alternative with the best payoff for the known event. The best alternative is the highest payoff if the payoffs are expressed as profits. If the payoffs are expressed as costs, the best alternative is the lowest payoff.

EXAMPLE A.5 Decisions under Certainty

A manager is deciding whether to build a small or a large facility. Much depends on the future demand that the facility must serve, and demand may be small or large. The manager knows with certainty the payoffs that will result under each alternative, shown in the following payoff table. The payoffs (in $000) are the present values of future revenues minus costs for each alternative in each event.

	POSSIBLE FUTURE DEMAND	
Alternative	**Low**	**High**
Small facility	200	270
Large facility	160	800
Do nothing	0	0

What is the best choice if future demand will be low?

SOLUTION

In this example, the best choice is the one with the highest payoff. If the manager knows that future demand will be low, the company should build a small facility and enjoy a payoff of $200,000. The larger facility has a payoff of only $160,000. The "do nothing" alternative is dominated by the other alternatives; that is, the outcome of one alternative is no better than the outcome of another alternative for each event. Because the "do nothing" alternative is dominated, the manager does not consider it further.

DECISION POINT

If management really knows future demand, it would build the small facility if demand will be low and the large facility if demand will be high. If demand is uncertain, it should consider other decision rules.

Decision Making under Uncertainty

Here, we assume that the manager can list the possible events but cannot estimate their probabilities. Perhaps, a lack of prior experience makes it difficult for the firm to estimate probabilities. In such a situation, the manager can use one of four decision rules:

1. *Maximin.* Choose the alternative that is the "best of the worst." This rule is for the *pessimist,* who anticipates the "worst case" for each alternative.
2. *Maximax.* Choose the alternative that is the "best of the best." This rule is for the *optimist* who has high expectations and prefers to "go for broke."
3. *Laplace.* Choose the alternative with the best *weighted payoff.* To find the weighted payoff, give equal importance (or, alternatively, equal probability) to each event. If there are n events, the importance (or probability) of each is $1/n$, so they add up to 1.0. This rule is for the *realist.*
4. *Minimax Regret.* Choose the alternative with the best "worst regret." Calculate a table of regrets (or opportunity losses), in which the rows represent the alternatives and the columns represent the events. A regret is the difference between a given payoff and the best payoff in the same column. For an event, it shows how much is lost by picking an alternative to the one that is best for this event. The regret can be lost profit or increased cost, depending on the situation.

EXAMPLE A.6 Decisions under Uncertainty

Reconsider the payoff matrix in Example A.5. What is the best alternative for each decision rule?

MyOMLab
Tutor A.4 in MyOMLab provides a new example to make decisions under uncertainty.

SOLUTION

a. *Maximin.* An alternative's worst payoff is the *lowest* number in its row of the payoff matrix, because the payoffs are profits. The worst payoffs ($000) are

Alternative	Worst Payoff
Small facility	200
Large facility	160

The best of these worst numbers is $200,000, so the pessimist would build a small facility.

b. *Maximax.* An alternative's best payoff ($000) is the *highest* number in its row of the payoff matrix, or

Alternative	Best Payoff
Small facility	270
Large facility	800

The best of these best numbers is $800,000, so the optimist would build a large facility.

c. *Laplace.* With two events, we assign each a probability of 0.5. Thus, the weighted payoffs ($000) are

Alternative	Weighted Payoff
Small facility	0.5(200) + 0.5(270) = **235**
Large facility	0.5(160) + 0.5(800) = **480**

The best of these weighted payoffs is $480,000, so the realist would build a large facility.

d. *Minimax Regret.* If demand turns out to be low, the best alternative is a small facility and its regret is 0 (or 200 – 200). If a large facility is built when demand turns out to be low, the regret is 40 (or 200 – 160).

	REGRET		
Alternative	**Low Demand**	**High Demand**	**Maximum Regret**
Small facility	200 – 200 = **0**	800 – 270 = **530**	**530**
Large facility	200 – 160 = **40**	800 – 800 = **0**	**40**

The column on the right shows the worst regret for each alternative. To minimize the maximum regret, pick a large facility. The biggest regret is associated with having only a small facility and high demand.

DECISION POINT

The pessimist would choose the small facility. The realist, optimist, and manager choosing to minimize the maximum regret would build the large facility.

Decision Making under Risk

Here we assume that the manager can list the events and estimate their probabilities. The manager has less information than with decision making under certainty, but more information than with decision making under uncertainty. For this intermediate situation, the *expected value* decision rule is widely used (both in practice and in this book). The expected value for an alternative is found by weighting each payoff with its associated probability and then adding the weighted payoff scores. The alternative with the best expected value (highest for profits and lowest for costs) is chosen.

This rule is much like the Laplace decision rule, except that the events are no longer assumed to be equally likely (or equally important). The expected value is what the *average* payoff would be if the decision could be repeated time after time. Of course, the expected value decision rule can result in a bad outcome if the wrong event occurs. However, it gives the best results if applied consistently over a long period of time. The rule should not be used if the manager is inclined to avoid risk.

EXAMPLE A.7 **Decisions under Risk**

Reconsider the payoff matrix in Example A.5. For the expected value decision rule, which is the best alternative if the probability of small demand is estimated to be 0.4 and the probability of large demand is estimated to be 0.6?

MyOMLab
Tutor A.5 in MyOMLab provides a new example to make decisions under risk.

SOLUTION

The expected value for each alternative is as follows:

Alternative	Expected Value
Small facility	0.4(200) + 0.6(270) = **242**
Large facility	0.4(160) + 0.6(800) = **544**

DECISION POINT

Management would choose a large facility if it used this expected value decision rule, because it provides the best long-term results if consistently applied over time.

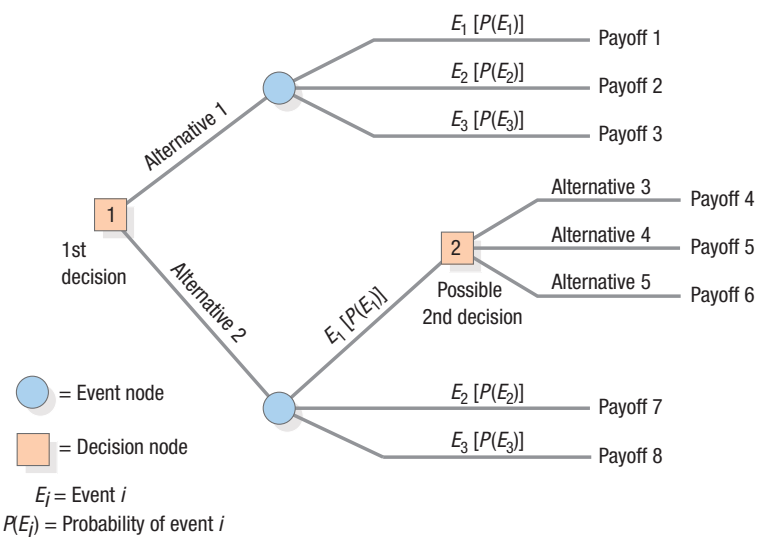
Decision Trees

The decision tree method is a general approach to a wide range of processes and supply chain decisions, such as product planning, process analysis, process capacity, and location. It is particularly valuable for evaluating different capacity expansion alternatives when demand is uncertain and sequential decisions are involved. For example, a company may expand a facility in 2013 only to discover in 2016 that demand is much higher than forecasted. In that case, a second decision may be necessary to determine whether to expand again or build a second facility.

A **decision tree** is a schematic model of alternatives available to the decision maker, along with their possible consequences. The name derives from the tree-like appearance of the model. It consists of a number of square *nodes*, representing decision points, which are left by *branches* (which should be read from left to right), representing the alternatives. Branches leaving circular, or chance, nodes represent the events. The probability of each chance event, $P(E)$, is shown above each branch. The probabilities for all branches leaving a chance node must sum to 1.0. The conditional payoff, which is the payoff for each possible alternative-event combination, is shown at the end of each combination. Payoffs are given only at the outset, before the analysis begins, for the end points of each alternative-event combination. In Figure A.4, for example, payoff 1 is the financial outcome the manager expects if alternative 1 is chosen and then chance event 1 occurs.

decision tree
A schematic model of alternatives available to the decision maker, along with their possible consequences.

No payoff can be associated yet with any branches farther to the left, such as alternative 1 as a whole, because it is followed by a chance event and is not an end point. Payoffs often are



◀ **FIGURE A.4**
A Decision Tree Model

expressed as the present value of net profits. If revenues are not affected by the decision, the payoff is expressed as net costs.

After drawing a decision tree, we solve it by working from right to left, calculating the *expected payoff* for each node as follows:

1. For an event node, we multiply the payoff of each event branch by the event's probability. We add these products to get the event node's expected payoff.
2. For a decision node, we pick the alternative that has the best expected payoff. If an alternative leads to an event node, its payoff is equal to that node's expected payoff (already calculated). We "saw off," or "prune," the other branches not chosen by marking two short lines through them. The decision node's expected payoff is the one associated with the single remaining unpruned branch. We continue this process until the leftmost decision node is reached. The unpruned branch extending from it is the best alternative to pursue. If multistage decisions are involved, we must await subsequent events before deciding what to do next. If new probability or payoff estimates are obtained, we repeat the process.

Various software is available for drawing decision trees. PowerPoint can be used to draw decision trees, although it does not have the capability to analyze the decision tree. More extensive capabilities, in addition to POM for Windows, are found with SmartDraw (**www.smartdraw.com**), PrecisionTree decision analysis from Palisade Corporation (**www.palisade.com**), and TreePlan (**www.treeplan.com/treeplan.htm**).

EXAMPLE A.8 Analyzing a Decision Tree

MyOMLab

Active Model A.3 in MyOMLab provides additional insight on this decision tree example and its extensions.

A retailer must decide whether to build a small or a large facility at a new location. Demand at the location can be either low or high, with probabilities estimated to be 0.4 and 0.6, respectively. If a small facility is built and demand proves to be high, the manager may choose not to expand (payoff = $223,000) or to expand (payoff = $270,000). If a small facility is built and demand is low, there is no reason to expand and the payoff is $200,000. If a large facility is built and demand proves to be low, the choice is to do nothing ($40,000) or to stimulate demand through local advertising. The response to advertising may be either modest or sizable, with their probabilities estimated to be 0.3 and 0.7, respectively. If it is modest, the payoff is estimated to be only $20,000; the payoff grows to $220,000 if the response is sizable. Finally, if a large facility is built and demand turns out to be high, the payoff is $800,000.

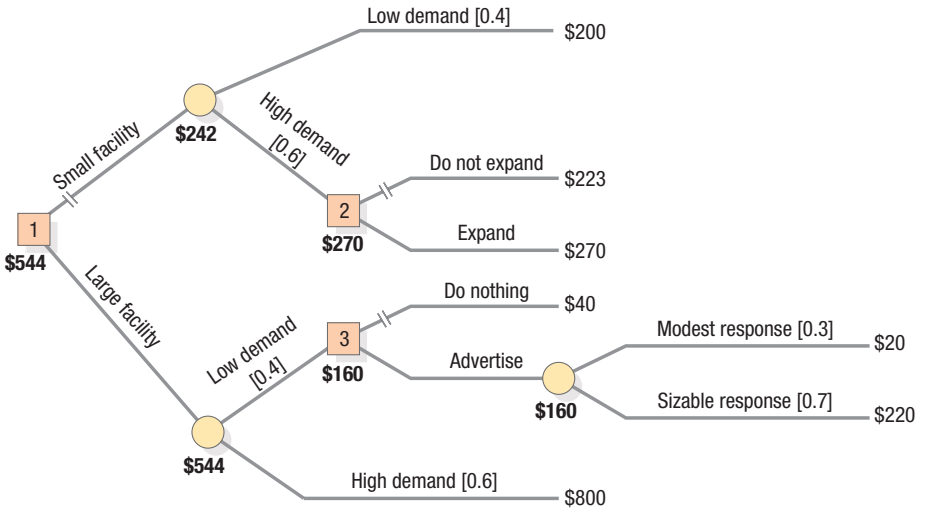
Draw a decision tree. Then analyze it to determine the expected payoff for each decision and event node. Which alternative—building a small facility or building a large facility—has the higher expected payoff?

SOLUTION

The decision tree in Figure A.5 shows the event probability and the payoff for each of the seven alternative-event combinations. The first decision is whether to build a small or a large facility. Its node is shown first, to the left, because it is the decision the retailer must make now. The second decision node—whether to expand at a later date—is reached only if a small facility is built and demand turns out to be high. Finally, the third decision point—whether to advertise—is reached only if the retailer builds a large facility and demand turns out to be low.

Analysis of the decision tree begins with calculation of the expected payoffs from right to left, shown on Figure A.5 beneath the appropriate event and decision nodes.

1. For the event node dealing with advertising, the expected payoff is 160, or the sum of each event's payoff weighted by its probability [0.3(20) + 0.7(220)].
2. The expected payoff for decision node 3 is 160 because *Advertise* (160) is better than *Do nothing* (40). Prune the *Do nothing* alternative.
3. The payoff for decision node 2 is 270 because *Expand* (270) is better than *Do not expand* (223). Prune *Do not expand*.
4. The expected payoff for the event node dealing with demand, assuming that a small facility is built, is 242 [or 0.4(200) + 0.6(270)].
5. The expected payoff for the event node dealing with demand, assuming that a large facility is built, is 544 [or 0.4(160) + 0.6(800)].
6. The expected payoff for decision node 1 is 544 because the large facility's expected payoff is largest. Prune *Small facility*.



◀ **FIGURE A.5**
Decision Tree for Retailer (in $000)

DECISION POINT

The retailer should build the large facility. This initial decision is the only one made now. Subsequent decisions are made after learning whether demand actually is low or high.

LEARNING GOALS IN REVIEW

1. **Explain break-even analysis, using both the graphic and algebraic approaches.** The section "Evaluating Services and Products," pp. 52–53, covers this analysis. Example A.1 and Solved Problem 4 demonstrate both approaches. Example A.3 shows its use in evaluating different processes.
2. **Define a preference matrix.** See the section "Preference Matrix," pp. 55–56, for making decisions involving unquantifiable factors, where some factors are rated more important than others. Example A.4 and Solved Problem 1 demonstrate the calculations.
3. **Explain how to construct a payoff table.** The section "Decision Theory," pp. 56–58, begins with the construction of a payoff table that shows the payoff for each feasible alternative and each event. See the table in Example A.5.
4. **Identify the maximin, maximax, Laplace, minimax regret, and expected value decision rules.** The sections "Decision Making under Uncertainty" and "Decision Making under Risk," pp. 57–58, cover these decision rules for when the outcomes associated with alternatives are in doubt. Examples A.6 and A.7 demonstrate how these rules work, and so does Solved Problem 2.
5. **Describe how to draw and analyze a decision tree.** The section "Decision Trees," pp. 59–61, show how to draw and analyze decision trees where several alternatives are available over time. Example A.8 and Solved Problem 3 shows how to work back from right to left, pruning as you go, until the best alternate is found for decision node 1.

MyOMLab helps you develop analytical skills and assess your progress with multiple problems on break-even analysis, sensitivity analysis, make-or-buy, the preference matrix, decisions under uncertainty, decisions under risks, and decision trees.

MyOMLab Resources	Titles	Link to the Book
Active Model Exercise	A.1 Break-Even Analysis A.3 Make-or-Buy Decision A.3 Decision Tree	Evaluating Services or Products; Examples A.1–A.2 (pp. 52–53); Solved Problem 4 (pp. 64–65) Evaluating Processes; Example A.3 (pp. 54–55) Decision Trees; Example A.8 (pp. 60–61)
OM Explorer Solvers	Break-Even Analysis Decision Theory Preference Matrix	Break-Even Analysis; Examples A.1–A.3 (pp. 52–55) Decision Theory; Example A.5–A.7 (pp. 57–59) Preference Matrix; Examples A.4–A.6 (pp. 55–57); Solved Problem 1 (pp. 62–63)

MyOMLab Resources	Titles	Link to the Book
OM Explorer Tutors	A.1 Break-Even, Evaluating Services and Products	Evaluating Services or Products; Example A.1–A.2 (pp. 52–53); Solved Problem 4 (pp. 64–65)
	A.2 Evaluating Processes	Evaluating Processes; Example A.3 (pp. 54–55)
	A.3 Preferences Matrix	Preference Matrix; Example A.4 (pp. 55–56)
	A.4 Decisions under Uncertainty	Decisions under Uncertainty; Example A.6 (pp. 57–58)
	A.5 Decisions under Risk	Decision Making under Risk; Example A.7 (p. 59); Solved Problem 2 (p. 63)
	A.6 Location Decisions under Uncertainty	Tutor A.6
POM for Windows	Decision Tables	Decision Theory
	Decision Trees (graphical)	Decision Trees (Graphical); Example A.8 (pp. 60–61); Solved Problem 3 (p. 64)
	Cost-Volume Analysis	Evaluating Processes
	Preference Matrix	Preference Matrix; Example A.4 (pp. 55–56); Solved Problem 1 (pp. 62–63)
	Break-Even Analysis	Break-Even Analysis; Example A.1–A.3 (pp. 52–55); Solved Problem 4 (pp. 64–65)
Virtual Tours	Flir infrared cameras	Decision Theory
	E* Trade investment services	Decision Theory
MyOMLab Supplements	F. Financial Analysis	MyOMLab Supplement F
Internet Exercises	Florida Small Business	Break–Even Analysis
Key Equations		
Image Library		

Key Equations

1. Break-even quantity: $Q = \dfrac{F}{p - c}$

2. Evaluating processes, make-or-buy indifference quantity: $Q = \dfrac{F_m - F_b}{c_b - c_m}$

Key Terms

break-even analysis 51
break-even quantity 51
decision theory 56
decision tree 59
fixed cost 52
payoff table 56
preference matrix 55
sensitivity analysis 53
variable cost 52

Solved Problem 1

Herron Company is screening three new product ideas: A, B, and C. Resource constraints allow only one of them to be commercialized. The performance criteria and ratings, on a scale of 1 (worst) to 10 (best), are shown in the following table. The Herron managers give equal weights to the performance criteria. Which is the best alternative, as indicated by the preference matrix method?

	RATING		
Performance Criterion	**Product A**	**Product B**	**Product C**
1. Demand uncertainty and project risk	3	9	2
2. Similarity to present products	7	8	6
3. Expected return on investment (ROI)	10	4	8
4. Compatibility with current manufacturing process	4	7	6
5. Competitive advantage	4	6	5

SOLUTION
Each of the five criteria receives a weight of 1/5 or 0.20.

Product	Calculation	Total Score
A	(0.20 × 3) + (0.20 × 7) + (0.20 × 10) + (0.20 × 4) + (0.20 × 4)	= 5.6
B	(0.20 × 9) + (0.20 × 8) + (0.20 × 4) + (0.20 × 7) + (0.20 × 6)	= 6.8
C	(0.20 × 2) + (0.20 × 6) + (0.20 × 8) + (0.20 × 6) + (0.20 × 5)	= 5.4

The best choice is product B. Products A and C are well behind in terms of total weighted score.

Solved Problem 2

MyOMLab
Tutor A.6 in MyOMLab examines decisions under uncertainty for a location example.

Adele Weiss manages the campus flower shop. Flowers must be ordered three days in advance from her supplier in Mexico. Although Valentine's Day is fast approaching, sales are almost entirely last-minute, impulse purchases. Advance sales are so small that Weiss has no way to estimate the probability of low (25 dozen), medium (60 dozen), or high (130 dozen) demand for red roses on the big day. She buys roses for $15 per dozen and sells them for $40 per dozen. Construct a payoff table. Which decision is indicated by each of the following decision criteria?

a. Maximin
b. Laplace
c. Maximax
d. Minimax regret

SOLUTION
The payoff table for this problem is

	DEMAND FOR RED ROSES		
Alternative	**Low (25 dozen)**	**Medium (60 dozen)**	**High (130 dozen)**
Order 25 dozen	$625	$625	$625
Order 60 dozen	$100	$1,500	$1,500
Order 130 dozen	($950)	$450	$3,250
Do nothing	$0	$0	$0

a. Under the minimax regret criteria, Weiss should order 130 dozen. The maximum regret of ordering 25 dozen occurs if demand is high: $3,250 – $625 = $2,625. The maximum regret of ordering 60 dozen occurs if demand is high: $3,250 – $1,500 = $1,750. The maximum regret of ordering 130 dozen occurs if demand is low: $625 – (–$950) = $1,575.

b. Under the maximax criteria, Weiss should order 130 dozen. The greatest possible payoff, $3,250, is associated with the largest order.

c. Under the Laplace criteria, Weiss should order 60 dozen. Equally weighted payoffs for ordering 25, 60, and 130 dozen are about $625, $1,033, and $917, respectively.

d. Under the maximin criteria, Weiss should order 25 dozen, because if demand is low, Weiss's profits are $625, the best of the worst payoffs.

Solved Problem 3

White Valley Ski Resort is planning the ski lift operation for its new ski resort. Management is trying to determine whether one or two lifts will be necessary; each lift can accommodate 250 people per day. Skiing normally occurs in the 14-week period from December to April, during which the lift will operate 7 days per week. The first lift will operate at 90 percent capacity if economic conditions are bad, the probability of which is believed to be about a 0.3. During normal times the first lift will be utilized at 100 percent capacity, and the excess crowd will provide 50 percent utilization of the second lift. The probability of normal times is 0.5. Finally, if times are really good, the probability of which is 0.2, the utilization of the second lift will increase to 90 percent. The equivalent annual cost of installing a new lift, recognizing the time value of money and the lift's economic life, is $50,000. The annual cost of installing two lifts is only $90,000 if both are purchased at the same time. If used at all, each lift costs $200,000 to operate, no matter how low or high its utilization rate. Lift tickets cost $20 per customer per day.

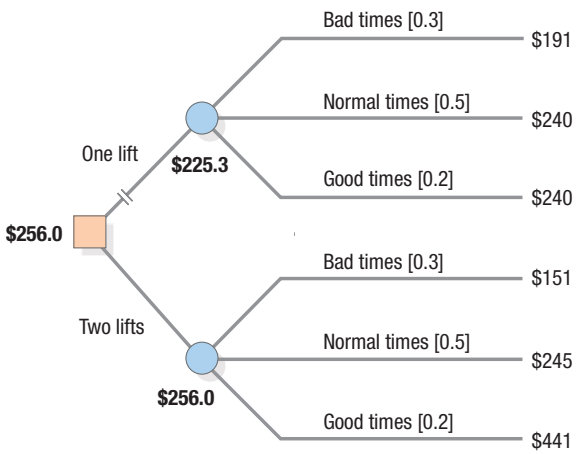
Should the resort purchase one lift or two?

SOLUTION

The decision tree is shown in Figure A.6. The payoff ($000) for each alternative-event branch is shown in the following table. The total revenues from one lift operating at 100 percent capacity are $490,000 (or 250 customers × 98 days × $20/customer-day).

Alternative	Economic Condition	Payoff Calculation (Revenue – Cost)
One lift	Bad times	0.9(490) – (50 + 200) = 191
	Normal times	1.0(490) – (50 + 200) = 240
	Good times	1.0(490) – (50 + 200) = 240
Two lifts	Bad times	0.9(490) – (90 + 200) = 151
	Normal times	1.5(490) – (90 + 400) = 245
	Good times	1.9(490) – (90 + 400) = 441

FIGURE A.6 ▶



Solved Problem 4

The owner of a small manufacturing business has patented a new device for washing dishes and cleaning dirty kitchen sinks. Before trying to commercialize the device and add it to his or her existing product line, the owner wants reasonable assurance of success. Variable costs are estimated at $7 per unit produced and sold. Fixed costs are about $56,000 per year.

a. Forecasted sales for the first year are 10,000 units if the price is reduced to $15. With this pricing strategy, what would be the product's total contribution to profits in the first year?

b. If the selling price is set at $25, how many units must be produced and sold to break even? Use both algebraic and graphic approaches.

SOLUTION

a. Beginning with the algebraic approach, we get

$$Q = \frac{F}{p - c} = \frac{56,000}{25 - 7}$$
$$= 3,111 \text{ units}$$

Using the graphic approach, shown in Figure A.7, we first draw two lines:

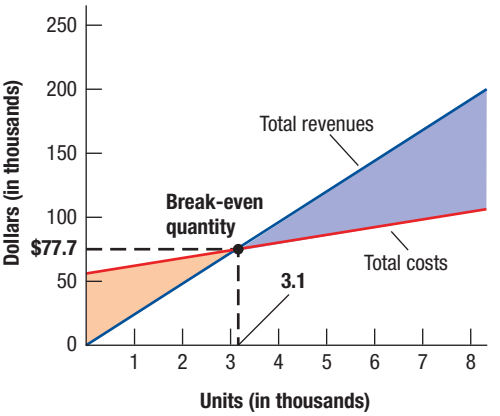
$$\text{Total revenue} = 25Q$$
$$\text{Total cost} = 56,000 + 7Q$$

The two lines intersect at $Q = 3,111$ units, the break-even quantity.

b. Total profit contribution = Total revenue – Total cost

$$= pQ - (F + cQ)$$
$$= 15(10,000) - [56,000 + 7(10,000)]$$
$$= \$24,000$$

◀ FIGURE A.7



Problems

The OM Explorer and POM for Windows software is available to all students using the 10th edition of this textbook. Go to **www.pearsonglobaleditions.com/krajewski** to download these computer packages. If you purchased MyOMLab, you also have access to Active Models software and significant help in doing the following problems. Check with your instructor on how best to use these resources. In many cases, the instructor wants you to understand how to do the calculations by hand. At the least, the software provides a check on your calculations. When calculations are particularly complex and the goal is interpreting the results in making decisions, the software entirely replaces the manual calculations.

BREAK-EVEN ANALYSIS

1. Tri-County G&T sells 140,000 MWh per year of electrical power to Boulder at $70 per MWh, has fixed costs of $83.2 million per year, and has variable costs of $30 per MWh. If Tri-County has 2,000,000 MWh of demand from its customers (other than Boulder), what will Tri-County have to charge to break even?

2. A news clipping service is considering modernization. Rather than manually clipping and photocopying articles of interest and mailing them to its clients, employees electronically input stories from most widely circulated publications into a database. Each new issue is searched for key words, such as a client's company name, competitors' names, type of business, and the company's products, services, and officers. When matches occur, affected clients are instantly notified via an online network. If the story is of interest, it is electronically transmitted, so the client often has the story and can prepare comments for follow-up interviews before the publication hits the street. The manual process has fixed costs of $400,000 per year and variable costs of $6.20 per clipping mailed. The price charged the client is $8.00 per clipping. The computerized process has fixed costs of $1,300,000 per year and variable costs of $2.25 per story electronically transmitted to the client.

a. If the same price is charged for either process, what is the annual volume beyond which the automated process is more attractive?

b. The present volume of business is 225,000 clippings per year. Many of the clippings sent with the current process are not of interest to the client or are multiple copies of the same story appearing in several publications. The news clipping service believes that

by improving service and by lowering the price to $4.00 per story, modernization will increase volume to 900,000 stories transmitted per year. Should the clipping service modernize?

c. If the forecasted increase in business is too optimistic, at what volume will the new process (with the $4.00 price) break even?

3. Spartan Castings must implement a manufacturing process that reduces the amount of particulates emitted into the atmosphere. Two processes have been identified that provide the same level of particulate reduction. The first process is expected to incur $350,000 of fixed cost and add $50 of variable cost to each casting Spartan produces. The second process has fixed costs of $150,000 and adds $90 of variable cost per casting.

a. What is the break-even quantity beyond which the first process is more attractive?

b. What is the difference in total cost if the quantity produced is 10,000?

4. Techno Corporation is currently manufacturing an item at variable costs of $4 per unit. Annual fixed costs of manufacturing this item are $141,000. The current selling price of the item is $11 per unit, and the annual sales volume is 35,000 units.

a. Techno can substantially improve the item's quality by installing new equipment at additional annual fixed costs of $65,000. Variable costs per unit would increase by $1, but, as more of the better-quality product could be sold, the annual volume would increase to 55,000 units. Should Techno buy the new equipment and maintain the current price of the item? Why or why not?

b. Alternatively, Techno could increase the selling price to $13 per unit. However, the annual sales volume would be limited to 45,000 units. Should Techno buy the new equipment and raise the price of the item? Why or why not?

5. An interactive television service that costs $11 per *month* to provide can be sold on the information highway for $15 per client per *month*. If a service area includes a potential of 18,000 customers, what is the most a company could spend on *annual* fixed costs to acquire and maintain the equipment?

6. A product at the Jennings Company enjoyed reasonable sales volumes, but its contributions to profits were disappointing. Last year, 17,500 units were produced and sold. The selling price is $22 per unit, the variable cost is $18 per unit, and the fixed cost is $80,000.

a. What is the percent change in the per-unit profit contribution generated by each alternative in part (d)?

b. Jennings is considering ways to either stimulate sales volume or decrease variable cost. Management believes that either sales can be increased by 30 percent or that variable cost can be reduced to 85 percent of its current level. Which alternative leads to higher contributions to profits, assuming that each is equally costly to implement? (Hint: Calculate profits for both alternatives and identify the one having the greatest profits.)

c. Jennings believes that a $1 reduction in price will increase sales by 50 percent. Is this enough for Jennings to break even? If not, by how much would sales have to increase?

d. If sales were not expected to increase, by how much would Jennings have to reduce their variable cost to break even?

e. What is the break-even quantity for this product? Use both graphic and algebraic approaches to get your answer.

7. The Tri-County Generation and Transmission Association is a nonprofit cooperative organization that provides electrical service to rural customers. Based on a faulty long-range demand forecast, Tri-County overbuilt its generation and distribution system. Tri-County now has much more capacity than it needs to serve its customers. Fixed costs, mostly debt service on investment in plant and equipment, are $89.2 million per year. Variable costs, mostly fossil fuel costs, are $29 per megawatt-hour (MWh, or million watts of power used for one hour). The new person in charge of demand forecasting prepared a short-range forecast for use in next year's budgeting process. That forecast calls for Tri-County customers to consume 2 million MWh of energy next year.

a. How much will Tri-County need to charge its customers per MWh to break even next year?

b. The Tri-County customers balk at that price and conserve electrical energy. Only 96 percent of forecasted demand materializes. What is the resulting surplus or loss for this nonprofit organization?

8. A restaurant is considering adding fresh brook trout to its menu. Customers would have the choice of catching their own trout from a simulated mountain stream or simply asking the waiter to net the trout for them. Operating the stream would require $9,600 in fixed costs per year. Variable costs are estimated to be $6.10 per trout. The firm wants to break even if 800 trout dinners are sold per year. What should be the price of the new item?

9. Hahn Manufacturing purchases a key component of one of its products from a local supplier. The current purchase price is $1,500 per unit. Efforts to standardize parts succeeded to the point that this same component can now be used in five different products. Annual component usage should increase from 150 to 750 units. Management wonders whether it is time to make the component in-house, rather than to continue buying it from the supplier. Fixed costs would increase by about $40,000 per year for the new equipment and tooling needed. The cost of raw materials and variable overhead would be about $1,100 per unit, and labor costs would be $300 per unit produced.

a. Should Hahn make rather than buy?

b. What is the break-even quantity?

c. What other considerations might be important?

10. Earthquake, drought, fire, economic famine, flood, and a pestilence of TV court reporters have caused an exodus from the City of Angels to Boulder, Colorado. The sudden increase in demand is straining the capacity of Boulder's electrical system. Boulder's alternatives have been

reduced to buying 300,000 MWh of electric power from Tri-County G&T at a price of $79 per MWh, or refurbishing and recommissioning the abandoned Pearl Street Power Station in downtown Boulder. Fixed costs of that project are $7 million per year, and variable costs would be $50 per MWh. Should Boulder build or buy?

11. Mary Williams, owner of Williams Products, is evaluating whether to introduce a new product line. After thinking through the production process and the costs of raw materials and new equipment, Williams estimates the variable costs of each unit produced and sold at $6 and the fixed costs per year at $60,000.

a. If the selling price is set at $12.50, Williams forecasts that first-year sales would increase to 15,000 units. Which pricing strategy ($14.00 or $12.50) would result in the greater total contribution to profits?

b. What other considerations would be crucial to the final decision about making and marketing the new product?

c. If the selling price is set at $18 each, how many units must be produced and sold for Williams to break even? Use both graphic and algebraic approaches to get your answer.

d. Williams forecasts sales of 10,000 units for the first year if the selling price is set at $14 each. What would be the total contribution to profits from this new product during the first year?

PREFERENCE MATRIX

12. The Forsite Company is screening three ideas for new services. Resource constraints allow only one idea to be commercialized at the present time. The following estimates have been made for the five performance criteria that management believes to be most important:

	RATING		
Performance Criterion	**Service A**	**Service B**	**Service C**
Capital equipment investment required	0.2	0.8	0.1
Expected return on investment (ROI)	0.2	0.8	0.7
Compatibility with current workforce skills	0.8	0.6	0.7
Competitive advantage	0.4	0.7	0.7
Compatibility with EPA requirements	0.6	0.8	0.4

a. Calculate a total weighted score for each alternative. Use a preference matrix and assume equal weights for each performance criterion. Which alternative is best? Worst?

b. Suppose that the expected ROI is given twice the weight assigned to each of the remaining criteria. (The sum of weights should remain the same as in part (a).) Does this modification affect the ranking of the three potential services?

13. Accel Express, Inc., collected the following information on where to locate a warehouse (1 = poor, 10 = excellent):

		LOCATION SCORE	
Location Factor	**Factor Weight**	**A**	**B**
Construction costs	10	1	2
Utilities available	10	8	6
Business services	10	3	7
Real estate cost	20	4	10
Quality of life	20	7	8
Transportation	30	3	10

a. Which location, A or B, should be chosen on the basis of the total weighted score?

b. If the factors were weighted equally, would the choice change?

14. You are in charge of analyzing five new suppliers of an important raw material and have been given the information shown below (1 = worst, 10 = best). Management has decided that criteria 2 and 3 are equally important and that criteria 1 and 4 are each four times as important as criterion 2. No more than 2 new suppliers are required but each new vendor must exceed a total score of 70 percent of the maximum total points to be considered.

	RATING				
Performance Criterion	**Vendor A**	**Vendor B**	**Vendor C**	**Vendor D**	**Vendor E**
Quality of raw material	8	7	3	6	9
Environmental impact	3	8	4	7	7
Responsiveness to order changes	9	5	7	6	5
Cost of raw material	7	6	9	2	7

a. Which new vendors do you recommend?

b. Would your decision change if the criteria were considered equally important?

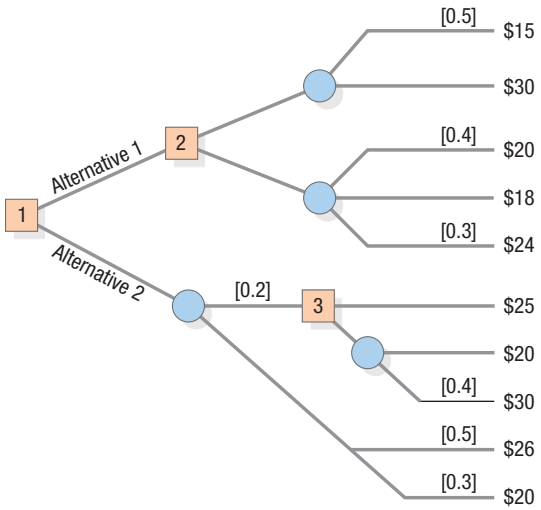
DECISION THEORY AND DECISION TREE

15. A manufacturing plant has reached full capacity. The company must build a second plant—either small or large—at a nearby location. The demand is likely to be high or low. The probability of low demand is 0.3. If demand is low, the large plant has a present value of $5 million and the small plant, a present value of $8 million. If demand is high, the large plant pays off with a present value of $18 million and the small plant with a present value of only $10 million. However, the small plant can be expanded later if demand proves to be high, for a present value of $14 million.

a. Draw a decision tree for this problem.

b. What should management do to achieve the highest expected payoff?

16. Analyze the decision tree in the following figure. What is the expected payoff for the best alternative? First, be sure to infer the missing probabilities.



17. A manager is trying to decide whether to buy one machine or two. If only one is purchased and demand proves to be excessive, the second machine can be purchased later. Some sales will be lost, however, because the lead time for producing this type of machine is six months. In addition, the cost per machine will be lower if both are purchased at the same time. The probability of low demand is estimated to be 0.30. The after-tax net present value of the benefits from purchasing the two machines together is $90,000 if demand is low and $180,000 if demand is high.

If one machine is purchased and demand is low, the net present value is $110,000. If demand is high, the manager has three options. Doing nothing has a net present value of $100,000; subcontracting, $140,000; and buying the second machine, $120,000.

a. Draw a decision tree for this problem.

b. How many machines should the company buy initially? What is the expected payoff for this alternative?

18. A manager is trying to decide whether to build a small, medium, or large facility. Demand can be low, average, or high, with the estimated probabilities being 0.25, 0.40, and 0.35, respectively.

A small facility is expected to earn an after-tax net present value of just $18,000 if demand is low. If demand is average, the small facility is expected to earn $75,000; it can be increased to medium size to earn a net present value of $60,000. If demand is high, the small facility is expected to earn $75,000 and can be expanded to medium size to earn $60,000 or to large size to earn $125,000.

A medium-sized facility is expected to lose an estimated $25,000 if demand is low and earn $140,000 if demand is average. If demand is high, the medium-sized facility is expected to earn a net present value of $150,000; it can be expanded to a large size for a net payoff of $145,000.

If a large facility is built and demand is high, earnings are expected to be $220,000. If demand is average for the large facility, the present value is expected to be $125,000; if demand is low, the facility is expected to lose $60,000.

a. Draw a decision tree for this problem.

b. What should management do to achieve the highest expected payoff?

c. Which alternative is best, according to each of the following decision criterion?

Maximin

Maximax

Minimax regret

19. Build-Rite Construction has received favorable publicity from guest appearances on a public TV home improvement program. Public TV programming decisions seem to be unpredictable, so Build-Rite cannot estimate the probability of continued benefits from its relationship with the show. Demand for home improvements next year may be either low or high. But Build-Rite must decide now whether to hire more employees, do nothing, or develop subcontracts with other home improvement contractors. Build-Rite has developed the following payoff table:

	DEMAND FOR HOME IMPROVEMENTS		
Alternative	**Low**	**Moderate**	**High**
Hire	($139,000)	$183,000	$448,000
Subcontract	$85,000	$158,000	$587,000
Do nothing	$100,000	$143,000	$411,000

Which alternative is best, according to each of the following decision criteria?

a. Maximin

b. Maximax

c. Laplace

d. Minimax Regret

20 . Benjamin Moses, chief engineer of Offshore Chemicals, Inc., must decide whether to build a new processing facility based on an experimental technology. If the new facility works, the company will realize a net profit of $20 million. If the new facility fails, the company will lose $10 million. Benjamin's best guess is that there is a 40 percent chance that the new facility will work.

What decision should Benjamin Moses make?

Selected References

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